



University of Nairobi

Department of Civil & Construction Engineering

COURSE FCC3 202:THEORY OF STRUCTURES IB

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Pre-requisites

1. FCC3 201 Analysis of Statically Determinate Structures IA
2. FCC3 231 Strength of Materials IA
3. FCC3 245 Material Science IA

Course Purpose

The course will provide learners with the ability to:

1. Analyse statically determinate structures (Beams, Frames and Trusses) and establish the displacements incurred at the salient points of the structure due to member straining.
2. Analyse statically determinate structures subject to moving loads, using the concept of Influence Lines

References

- Onsongo W. M. (1993) Analysis of Statically Determinate Structures. University of Nairobi Press
- 2. Coates R. C, Coutie M. G, Kong, F. K. (1997) Structural Analysis. Chapman and Hall, 3rd Ed
- 3. N.C Sinha, (2011), Elements of Structural Analysis. New Central Book Agency (P) Ltd, London, 1st Ed

Expected Learning Outcomes

At the end of this course, the learner should be able to: -

1. Analyse loaded Beams to determine both translational and rotational displacements using the direct integration/Macaulay's method, moment area method, conjugate beam method and virtual force method.
2. Analyse loaded Pin-jointed frames (trusses), to determine translational displacements using Energy methods (Real work approach, Castigliano's 1st Theorem and Virtual force method).
3. Analyse statically determinate Beams subjected to moving loads, using the concept of Influence Lines. Sketch the influence line diagrams (ILD) for reactions, shear forces and bending moments at given points along the beam. Use the ILD to determine reactions, shear forces and bending moments at given points when load is located at another point along the beam. Draw the maximum shear force and bending moment envelopes and use it to determine maximum shear force and bending moment at any given point on the beam.
4. Analyse statically determinate Pin-jointed frames (trusses) subjected to moving loads, using the concept of Influence Lines. Sketch the influence line diagrams (ILD) for reactions and member internal forces. Use the ILD to determine reactions and member internal forces when load is located at different points of the truss.

COURSE CONTENT

1. **Deflection of Beams** by direct integration method, moment area method, conjugates beam method and virtual force method.
2. **Truss displacements** by virtual force and graphical methods.
3. **Influence lines for statically determinate structures**

1. INTRODUCTION.

1.1. SCOPE

The scope of FCC3 202 includes the determination of the displacements induced in: -

1. Beams and Rigid-jointed frames
2. Pin-jointed frames (trusses).
3. Hybrid frames and Arches.

➤ The term **displacement** is used in the general sense to mean either translational displacement (called deflection), or rotational displacement (called slope).

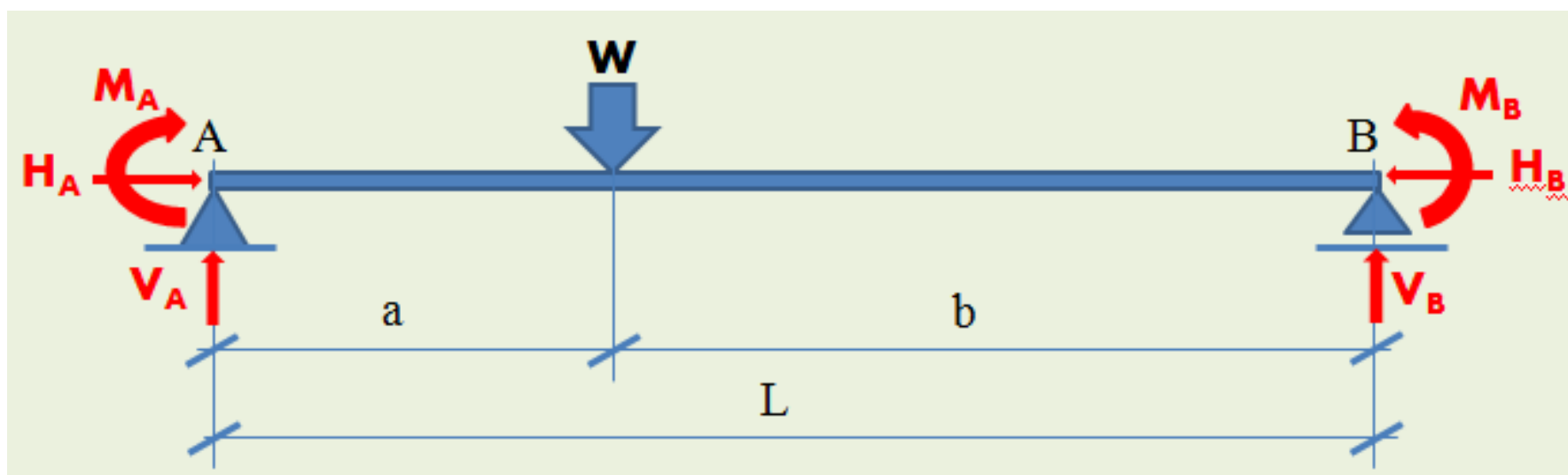
➤ Similarly the term **force** is used in the general sense to mean either a linear force F , or a bending moment M .

➤ **Moving Loads:** In FCC3 201, the applied loads were acting at specified points on the structure. In certain structures (such as bridges, crane supports etc) , the loads move along the structure. In FCC3 202, we shall analyze determinate structures subject to moving loads, using the concept of Influence Lines.

1. INTRODUCTION.

1.2. Types of structural problems.

(i). Fixed force magnitude, fixed force position.

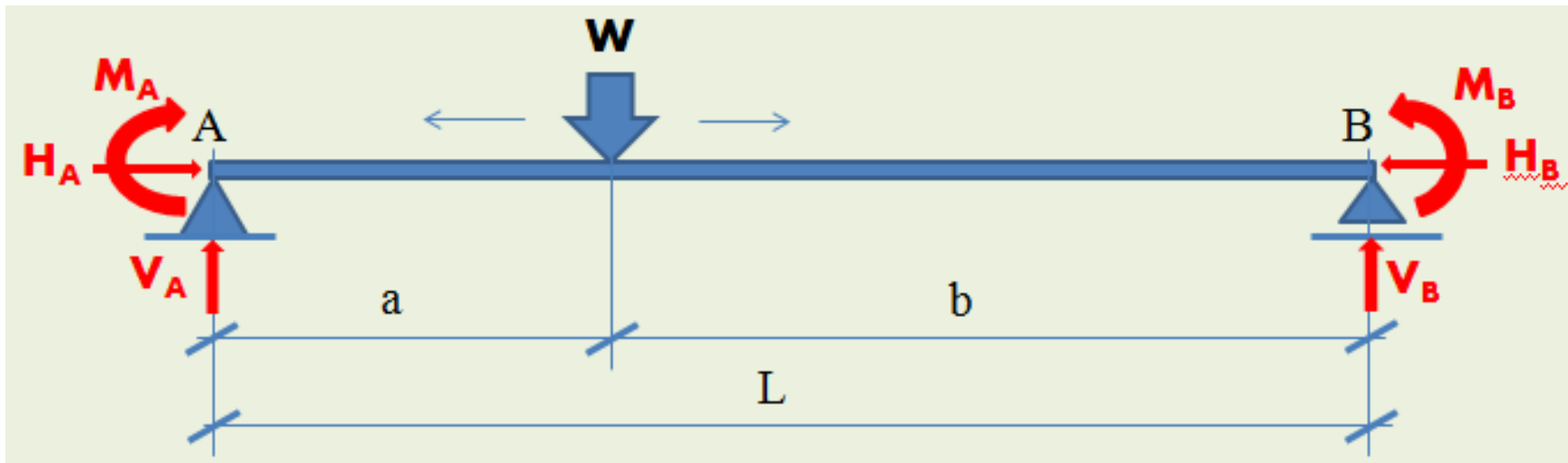


Analysis by statical methods:

$$\Sigma H = 0, \quad \Sigma V = 0, \quad \Sigma M = 0$$

1. INTRODUCTION.

(ii). Fixed force magnitude, changing force position.

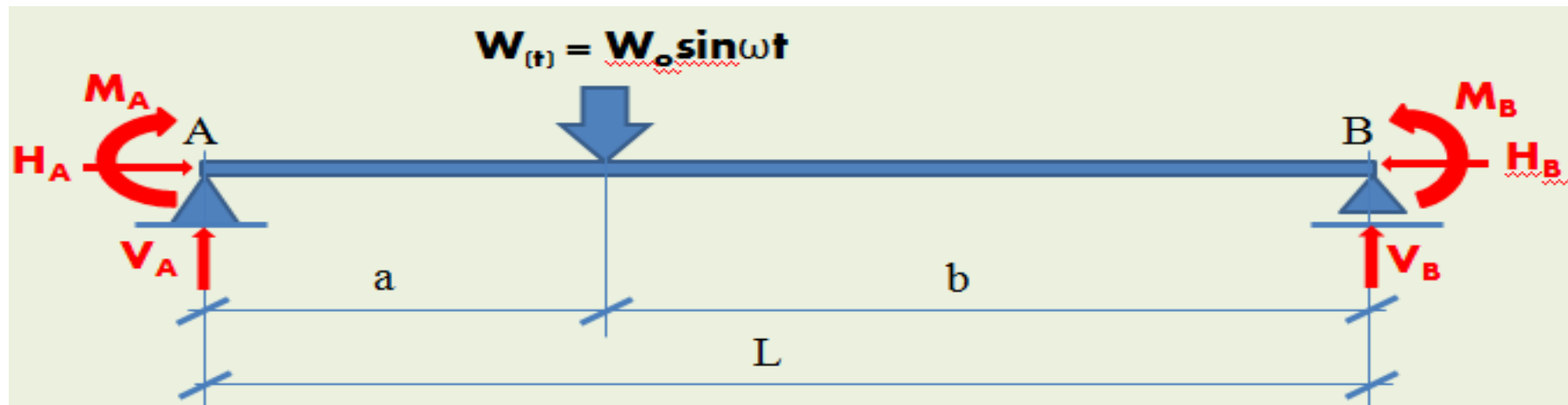


Analysis by influence lines.

Maximum bending moment and shear force envelopes.

1. INTRODUCTION.

(iii). Changing force magnitude, fixed force position.



Analysis by dynamic methods. Of interest are:

- Natural frequency of vibrating system.
- Frequency (rate) of vibration.
- Amplitude (amount of sway from position of rest) of vibration, etc.

1. INTRODUCTION.

1.3 Beam Bending Theory (Euler - Bernoulli Equation)

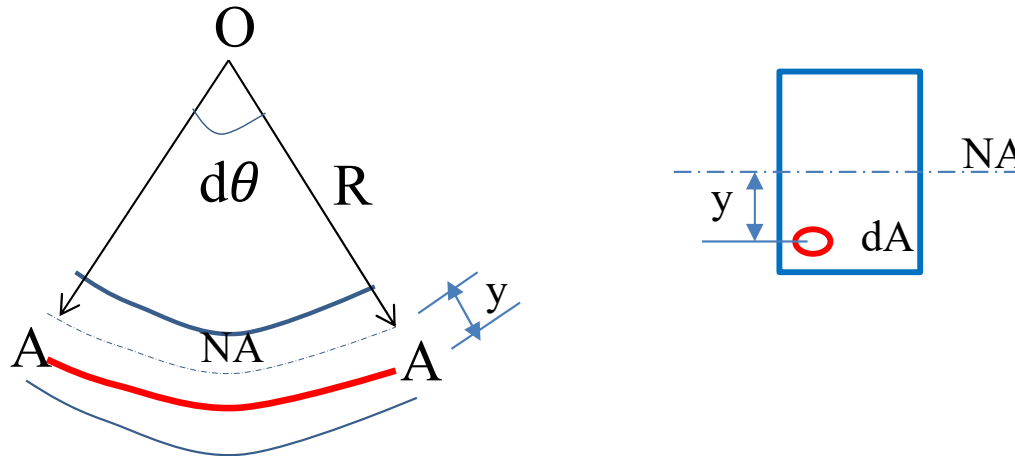
In classical **beam deflection theory** the following assumptions are made: -

1. The beam is initially straight
2. The radius of curvature is large compared to the cross-sectional dimensions of the beam
3. The material is homogeneous, isotropic, and obeys Hooke's law.
4. The stresses are uniform across the width and are within the elastic limit.
5. Cross section of the beam is symmetrical about the plane of bending
6. Plane sections before straining remain plane after straining.
7. Every longitudinal fiber is free to extend or contract without being restrained by its neighbor

Consider a small element of a beam under flexure as shown below:

Let R be the radius of curvature (measured from the centre of curvature O , to the neutral axis), and $d\theta$ be the change in slope between the ends of the considered element.

1. INTRODUCTION.



Consider fibers AA at distance y from the neutral axis

Original length of fibers AA = $Rd\theta$

Final length of fibers AA = $(R + y)d\theta$

$$\text{Flexural strain} = \frac{[(R+y)d\theta - Rd\theta]}{Rd\theta}$$

$$\text{Thus } \varepsilon = y/R$$

The flexural stress is given by $\sigma = E\varepsilon$

$$\text{Thus } \sigma = Ey/R$$

Hence we can write

$$\sigma/y = E/R \text{ ----- Eqn. 1.1}$$

1. INTRODUCTION.

Equilibrium of moments requires that the applied moment be equal and opposite to the moment of resistance of the section, thus

Force in the small element of area, $dF = \sigma \cdot dA$

Moment in the small element of area, $dM = y \cdot \sigma \cdot dA$

Total moment of resistance of section

$$M = \int y \cdot \sigma \cdot dA = \int y \cdot \left(\frac{Ey}{R} \right) \cdot dA = (E/R) \int y^2 dA$$

But $\int y^2 dA = I$, the second moment of area of the section Thus $M = (E/R) I$

Hence we can write

$$\mathbf{M/I = E/R \text{ ----- Eqn. 1.2}}$$

Equations (1) and (2) above are combined to form the classical Euler - Bernoulli equation expressed as: $\sigma/y = E/R = M/I$

$$\text{Which can also be expressed as: } \mathbf{M/EI = 1/R \text{ ----- Eqn.1.3}}$$

The quantity EI is the flexural stiffness of the beam, while the quantity $1/R$ is the curvature of the beam. R is the radius of curvature.

1. INTRODUCTION.

1.4 Fundamental Equation of Beams Based on Deflection

The standard equation of curvature of a beam can be expressed mathematically in terms of deflection (y) as follows:

$$\frac{1}{R} = \frac{\frac{d^2y}{dx^2}}{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{\frac{3}{2}}}$$

where y is the deflection of the beam at a horizontal distance x

From limits, $\left(\frac{dy}{dx}\right)^2$ is very small and hence negligible. Therefore $\frac{1}{R} = \frac{d^2y}{dx^2}$

Or $\frac{M}{EI} = \frac{d^2y}{dx^2} \rightarrow$ Fundamental equation of beam based on deflection.

The internal moment M_r acts to balance the applied moment (M)

1. INTRODUCTION.

PROOF

Let us consider a beam subjected to pure bending. Unless it is a very deep beam with a short span, deflection due to shear is negligible, hence effect of shear is not considered.

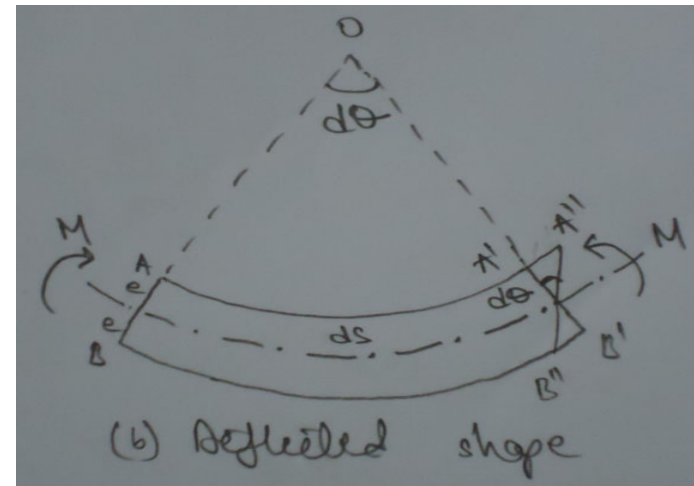
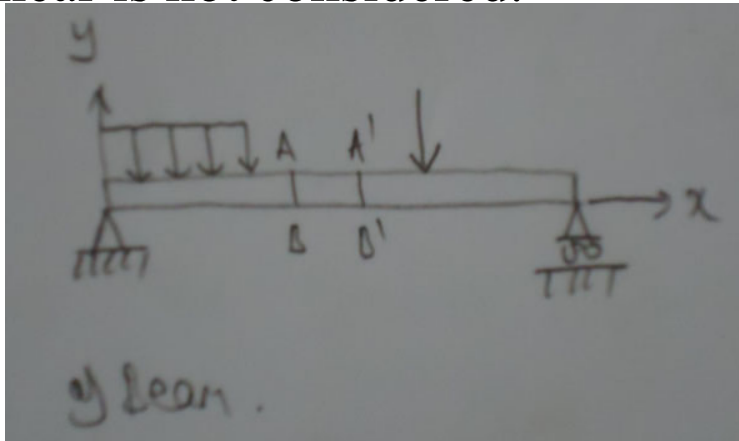


Figure above shows the bent shape of a small deflected element of the beam. Two originally plane parallel sections ds distance apart AB and $A'B'$ will now make a small angle $d\theta$. $A''B''$ is drawn parallel to AB . From Fig. (b), it is seen that: $A'A'' = B'B'' = ed\theta$

If σ is the maximum stress in the extreme top and bottom fibers, E is the Young's modulus of elasticity, strain $\epsilon = \sigma/E$ and extension or contraction

over length ds , $\Delta L = \frac{\sigma}{E} \cdot ds$

Hence, $ed\theta = \frac{\sigma}{Es} \cdot ds$ or $\frac{d\theta}{ds} = \frac{\sigma}{eE}$

1. INTRODUCTION.

Using the familiar strength of materials formula

$$\frac{M}{I} = \frac{\sigma}{y} \quad \text{Or} \quad \sigma = \frac{My}{I}$$

Here, $y = e$. $\therefore \frac{d\theta}{ds} = \frac{Me}{I} \cdot \frac{1}{eE} = \frac{M}{EI} \dots\dots\dots \text{Eqn. 1.4}$

$\frac{d\theta}{ds}$ Is called curvature and is defined as the rate at which the curve is changing direction.

Consider:

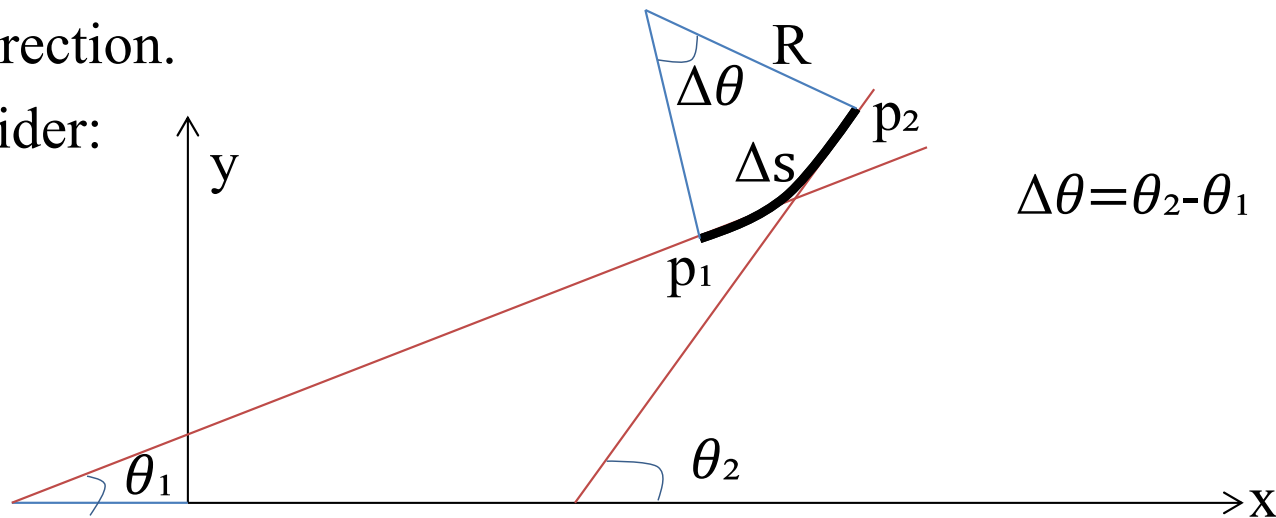


Figure 2: Curvature Vs x-y plane

Let P₁ and P₂ be two points on a curve a small distance Δs apart. The change of direction from P₁ to P₂ = θ₂ - θ₁ = Δθ

1. INTRODUCTION.

The average rate of change of direction between P_1 and $P_2 = \frac{\Delta\theta}{\Delta s}$. The limiting value of this ratio as Δs approaches zero is called the curvature and the radius of curvature reciprocal to this curvature. Hence curvature $p = \frac{1}{R}$

$\lim \frac{\Delta\theta}{\Delta s} = \frac{d\theta}{ds}$ To convert $\frac{d\theta}{ds}$ to x,y coordinates as shown in Fig.2

$$\tan \theta = \frac{dy}{dx} \text{ and } \frac{d}{dx} (\tan \theta) = \frac{d^2y}{dx^2}$$

$$\text{or } \frac{d}{d\theta} (\tan \theta) \frac{d\theta}{dx} = \frac{d^2y}{dx^2} \quad \text{or } \frac{\cos^2\theta + \sin^2\theta}{\cos^2} \cdot \frac{d\theta}{dx} = \frac{d^2y}{dx^2}$$

$$\text{or } (1 + \tan^2 \theta) \frac{d\theta}{dx} = \frac{d^2y}{dx^2}$$

$$\frac{d\theta}{dx} = \frac{\frac{d^2y}{dx^2}}{1 + \tan^2 \theta} = \frac{\frac{d^2y}{dx^2}}{1 + \left(\frac{dy}{dx}\right)^2}$$

$$\text{Now } \frac{d\theta}{ds} = \frac{d\theta}{dx} \cdot \frac{dx}{ds} = \frac{\frac{d^2y}{dx^2}}{1 + \left(\frac{dy}{dx}\right)^2} \cdot \frac{dx}{\sqrt{(dx^2 + dy^2)}}$$

1. INTRODUCTION.

$$\text{Or } \frac{d\theta}{ds} = \frac{\frac{d^2y}{dx^2}}{1 + \left(\frac{dy}{dx}\right)^2} \cdot \frac{1}{\sqrt{\frac{dx^2 + dy^2}{dx^2}}} = \frac{\frac{d^2y}{dx^2}}{1 + \left(\frac{dy}{dx}\right)^2} \cdot \frac{1}{\sqrt{1 + \left(\frac{dy}{dx}\right)^2}} = \frac{\frac{d^2y}{dx^2}}{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{\frac{3}{2}}} \dots \text{Eqn.1.5}$$

$$\text{Equating Equation 1.3, 1.4 \& 1.5, } \frac{1}{R} = \frac{d\theta}{ds} = \frac{M}{EI} = \frac{\frac{d^2y}{dx^2}}{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{\frac{3}{2}}}$$

Usually, the elastic deflection of a beam is small. dy/dx is then small and $\left(\frac{dy}{dx}\right)^2$ is extremely smaller and can be neglected in comparison to 1.

$$\text{Hence, } \frac{M}{EI} \approx \frac{d^2y}{dx^2}$$

The derivation is based on coordinate axes as shown. However, usually, y axis is taken as positive downwards instead of upwards. In that case, the equation will be:-

$$\frac{M}{EI} \approx -\frac{d^2y}{dx^2} \quad \text{Eqn. 1.6}$$

The solution of the above differential equation of the deflection curve involves integrating twice and hence the method is known as double integration method.

2. THE DIRECT INTEGRATION METHOD

Curvature, Slope and Deflection

The **direct integration method** (proposed by Daniel Bernoulli © 1750) is **based on the** assumption that EI is constant, so that the beam curvature can be expressed using Eqn.1.6

Curvature: $EI \frac{d^2y}{dx^2} = -M$

By integrating twice we obtain:

1. Slope: $EI \frac{dy}{dx} = \int -M \cdot dx + C1$

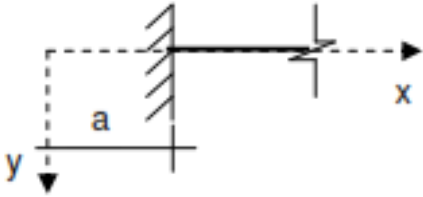
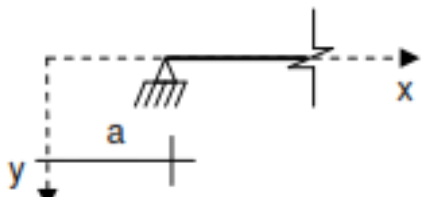
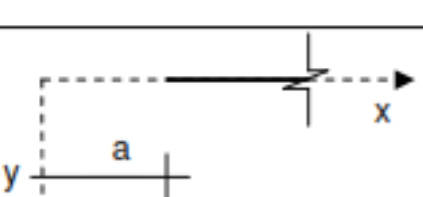
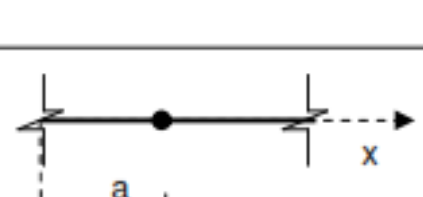
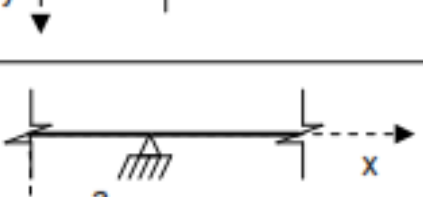
2. Deflection: $EI y = \int \{ \int -M \cdot dx + C1 \} dx + C2$

NOTE

The moment M is expressed as a function of x , and The constants of integration $C1$ and $C2$ are obtained from the existing boundary conditions.

2. THE DIRECT INTEGRATION METHOD

The various boundary conditions are summarized in the table below.

Fixed Support		Deflection = 0 Slope = 0	$y _a = 0$ $\frac{dy}{dx} _a = 0$
Pin support or Roller support at end		Deflection = 0 Moment = 0	$y _a = 0$ $\frac{d^2y}{dx^2} _a = 0$
Free end		Moment = 0 Shear = 0	$\frac{d^2y}{dx^2} _a = 0$ $\frac{d^3y}{dx^3} _a = 0$
Internal hinge		Moment = 0	$\frac{d^2y}{dx^2} _a = 0$
Continuous Support		Deflection = 0	$y _a = 0$

2. THE DIRECT INTEGRATION METHOD

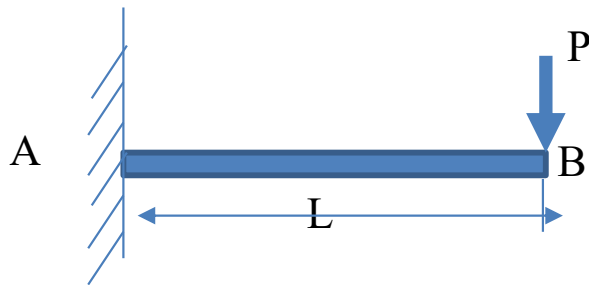
Solution Procedure

1. Determine the support reaction components to obtain a FBD of the given beam.
2. For each region:
 - i. Consider a FBD from the LH end to some point, distance x , within that region.
 - ii. Express the bending moment M in terms of x
 - iii. Use Eqn1.6 to express $EI \frac{d^2y}{dx^2} = -M$
 - iv. Integrate twice, assuming $EI = \text{constant}$.
3. Next region
4. Use the existing boundary conditions to solve for the constants of integration.
5. Obtain mathematical expressions for the deflection profile and the slope profile.
6. Use these expressions to evaluate the required displacements.

2. THE DIRECT INTEGRATION METHOD

Worked Example 2.1

Determine the deflection δ_B and slope θ_B at the free end B of the cantilever beam below, in terms of P, L and EI. Assume $EI = \text{constant}$.



Solution



Support Reactions

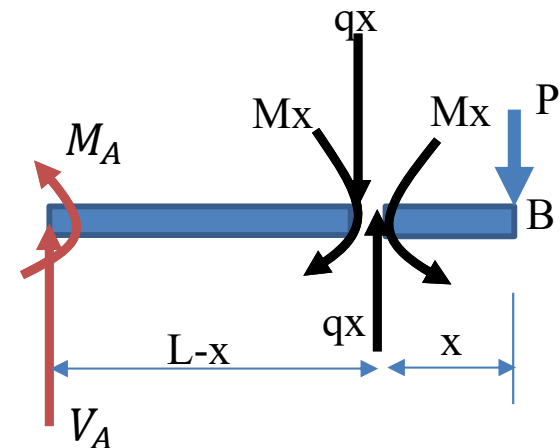
$$\sum M \text{ at } A = 0$$

$$\Rightarrow -M_A + PL = 0$$

$$\Rightarrow M_A = PL$$

And

$$\sum F_v = 0 \Rightarrow V_A = P$$



$$M_x = -Px$$

$$EI \frac{d^2 y}{dx^2} = -M \text{ or } EI \frac{d^2 y}{dx^2} = -(-Px)$$

$$\text{or } EI \frac{d^2 y}{dx^2} = Px$$

$$EI \frac{dy}{dx} = \frac{Px^2}{2} + C_1$$

$$EI Y = \frac{Px^3}{6} + C_1 x + C_2$$

2. THE DIRECT INTEGRATION METHOD

Solution

Apply Boundary conditions,

At $x = l, y = 0$ and $\frac{dy}{dx} = 0$

$$\text{Implies } 0 = \frac{Pl^2}{2} + C_1 \Rightarrow C_1 = -\frac{Pl^2}{2}$$

$$0 = \frac{Pl^3}{6} + C_1 l + C_2 \Rightarrow C_2 = \frac{Pl^3}{3}$$

Deflection and slope profiles

$$\therefore EI \frac{dy}{dx} = \frac{Px^2}{2} - \frac{Pl^2}{2} = \text{slope profile}$$

and

$$EI Y = \frac{Px^3}{6} - \frac{Pl^2}{2}x + \frac{Pl^3}{3} = \text{deflection profile}$$

Deflection and Slope at the free end B,

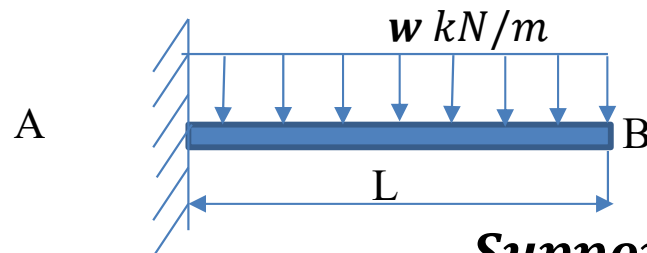
at end B, $x = 0$

$$\therefore \text{Slope } \theta_B = -\frac{Pl^2}{2EI} \text{ and Deflection } \delta_B = \frac{Pl^3}{3EI}$$

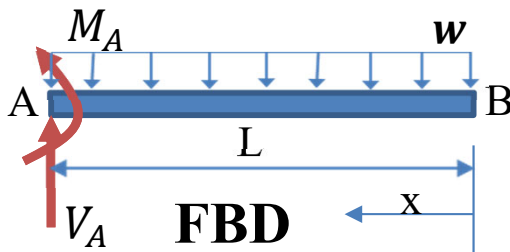
2. THE DIRECT INTEGRATION METHOD

Worked Example 2.2

Determine the deflection δ_B and slope θ_B at the free end B of the cantilever beam below, in terms of w , L and EI . Assume $EI = \text{constant}$.



Solution



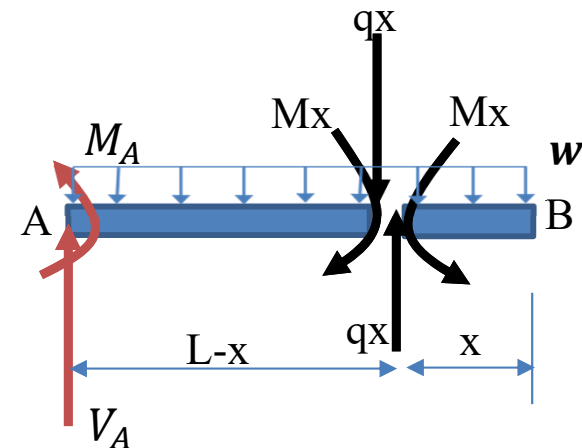
Support Reactions

$$\begin{aligned}\sum M \text{ at } A &= 0 \\ \Rightarrow -M_A + \frac{wL^2}{2} &= 0 \\ \Rightarrow M_A &= \frac{wL^2}{2} \\ \sum F_v &= 0 \Rightarrow V_A = wL\end{aligned}$$

$$M_x = -wx \cdot \frac{x}{2} = -\frac{wx^2}{2}$$

$$EI \frac{d^2y}{dx^2} = -M \quad \text{or} \quad EI \frac{d^2y}{dx^2} = -\left(-\frac{wx^2}{2}\right)$$

$$\text{or } EI \frac{d^2y}{dx^2} = \frac{wx^2}{2}$$



$$EI \frac{dy}{dx} = \frac{wx^3}{6} + C_1$$

$$EI Y = \frac{wx^4}{24} + C_1x + C_2$$

2. THE DIRECT INTEGRATION METHOD

Solution

Apply Boundary conditions,

At $x = l, y = 0$ and $\frac{dy}{dx} = 0$

$$\text{Implies } 0 = \frac{wl^3}{6} + C_1 \Rightarrow C_1 = -\frac{wl^3}{6}$$

$$\text{and } 0 = \frac{wl^4}{24} + \left(-\frac{wl^3}{6}\right)l + C_2 \Rightarrow C_2 = \frac{wl^4}{8}$$

Deflection and Slope profiles

$$\therefore EI \frac{dy}{dx} = \frac{wx^3}{6} - \frac{wl^3}{6} = \text{slope profile}$$

and

$$EI Y = \frac{wx^4}{24} - \frac{wl^3}{6}x + \frac{wl^4}{8} = \text{deflection profile}$$

Deflection and Slope at B

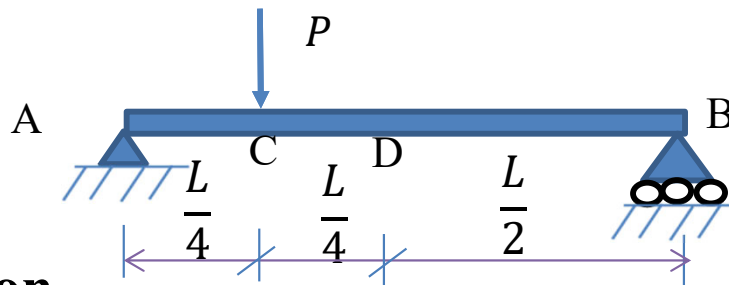
At the free end B, $x=0$

$$\therefore \text{Slope } \theta_B = -\frac{wl^3}{6EI} \text{ and Deflection } \delta_B = \frac{wl^4}{8EI}$$

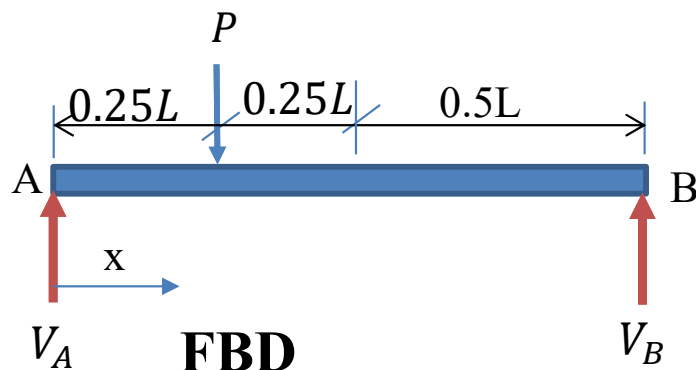
2. THE DIRECT INTEGRATION METHOD

Worked Example 2.3

Determine the deflection at under the point load and at mid-span and slopes θ_A & θ_B at the supports of the simply supported beam below, in terms of P , L and EI . Assume $EI = \text{constant}$.



Solution



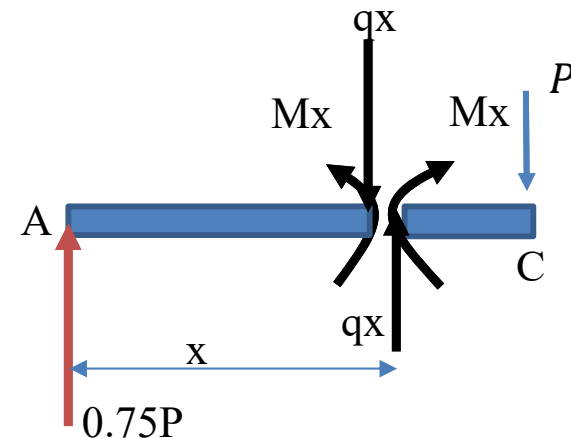
Support Reactions

$$\sum F_v = 0 \Rightarrow V_A + V_B = P$$

And

$$\sum M \text{ at } A = 0 \Rightarrow V_B \cdot L = 0.25LP$$

$$\Rightarrow V_B = 0.25P \text{ and } V_A = 0.75P$$



Section AC

$$M_x = \frac{3Px}{4}$$

$$EI \frac{d^2y}{dx^2} = -M$$

$$\text{or } EI \frac{d^2y}{dx^2} = -\frac{3Px}{4}$$

$$EI \frac{dy}{dx} = -\frac{3Px^2}{8} + C_1$$

$$EI Y = -\frac{3Px^3}{24} + C_1x + C_2$$

2. THE DIRECT INTEGRATION METHOD

Solution

Section CB

$$M_x = \frac{P(L-x)}{4}$$

$$EI \frac{d^2 y}{dx^2} = -M$$

$$\text{or } EI \frac{d^2 y}{dx^2} = -\frac{P(L-x)}{4}$$

$$EI \frac{dy}{dx} = \frac{Px^2}{8} - \frac{PLx}{4} + C_3$$

$$EI Y = \frac{Px^3}{24} - \frac{PLx^2}{8} + C_3x + C_4$$

Apply Boundary conditions,

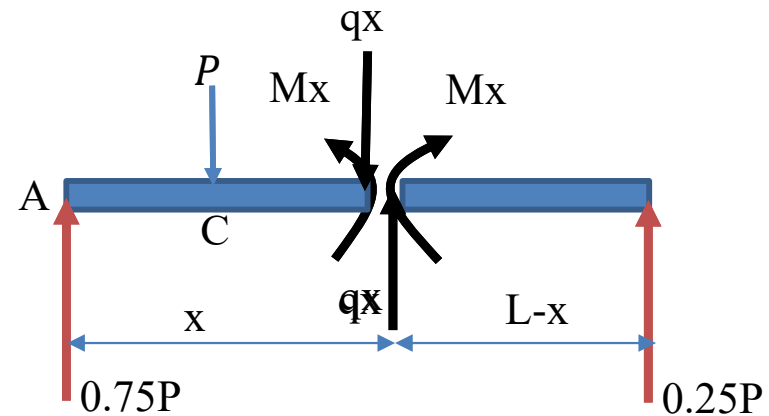
1. Section AC, at $x = 0, y = 0$

2. Section CB at $x = L, y = 0$

3. Continuity at C, $x = \frac{L}{4}, Y_{AC} = Y_{CB}$

4. Continuity at C, $x = \frac{L}{4},$

$$EI \frac{dy}{dx}_{AC} = EI \frac{dy}{dx}_{CB}$$



Implies $0 = 0 + 0 + C_2 \Rightarrow C_2 = 0 \dots (i)$

$$\text{Also } 0 = \frac{PL^3}{24} - \frac{PL^3}{8} + C_3L + C_4$$

$$\text{Or } 0 = -\frac{PL^3}{12} + C_3L + C_4 \dots (ii)$$

$$\text{At } x = \frac{l}{4},$$

$$-\frac{3P(\frac{l}{4})^3}{24} + \frac{C_1L}{4} = \frac{P(\frac{l}{4})^3}{24} - \frac{PL(\frac{l}{4})^2}{8} + \frac{L}{4} \cdot C_3 + C_4$$

Or

$$-\frac{PL^3}{512} + \frac{C_1L}{4} = \frac{PL^3}{1536} - \frac{PL^3}{128} + \frac{L}{4} \cdot C_3 + C_4 \dots (iii)$$

2. THE DIRECT INTEGRATION METHOD

$$\text{And } -\frac{3P(\frac{L}{4})^2}{8} + C_1 = \frac{P(\frac{L}{4})^2}{8} - \frac{PL(\frac{L}{4})}{4} + C_3$$

$$\text{Or } -\frac{3PL^2}{128} + C_1 = \frac{PL^2}{128} - \frac{PL^2}{16} + C_3 \dots\dots\dots (iv)$$

Solving Eqns. i) to iv) simultaneously we get

$$C_1 = \frac{7PL^2}{128}, C_2 = 0, C_3 = \frac{11PL^2}{128}, C_4 = -\frac{PL^3}{384}$$

Deflection and Slope profiles section AC

$$EI \frac{dy}{dx} = -\frac{3Px^2}{8} + \frac{7PL^2}{128} \Rightarrow \text{slope profile}$$

$$EI Y = -\frac{3Px^3}{24} + \frac{7PL^2x}{128} \Rightarrow \text{deflection profile}$$

Deflection and Slope profiles section CB

$$EI \frac{dy}{dx} = \frac{Px^2}{8} - \frac{PLx}{4} + \frac{11PL^2}{128}$$

$$EI Y = \frac{Px^3}{24} - \frac{PLx^2}{8} + \frac{11PL^2x}{128} - \frac{PL^3}{384}$$

Slope at A

use eqns. for section AC

$$, x = 0 \therefore \text{Slope } \theta_A = \frac{7PL^2}{128EI}$$

Deflection at C

use eqns. for section AC or CB

$$, x = L/4$$

$$\therefore \text{Deflection } \delta_C = \frac{3PL^3}{256EI}$$

Slope at B

use eqns. for section CB

$$, x = L \therefore \text{Slope } \theta_B = -\frac{5PL^2}{128EI}$$

Deflection at Mid-span, D

use eqns. for section CB

Where mid-span lies

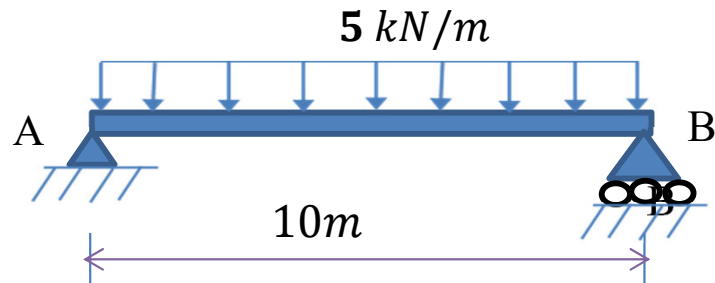
$$, x = L/2$$

$$\therefore \text{Deflection } \delta_D = \frac{11PL^3}{768EI}$$

2. THE DIRECT INTEGRATION METHOD

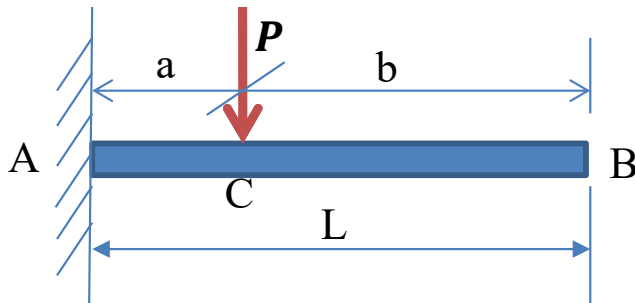
Exercise 2.1

Determine the deflection at mid-span and slopes θ_A & θ_B at the supports of the simply supported beam below. Assume $EI = \text{constant}$.



Exercise 2.2

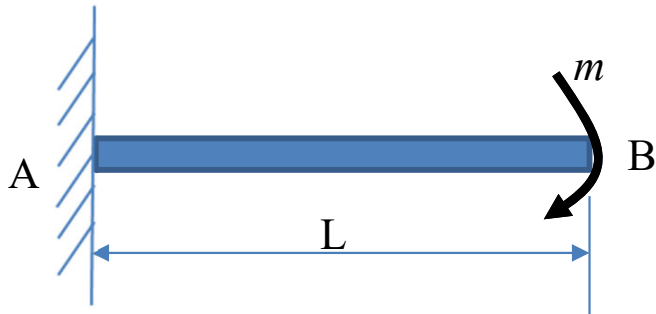
Determine the deflection at C & B and slopes θ_C & θ_B for beam below, in terms of P, L, a, b & EI . Assume $EI = \text{constant}$.



2. THE DIRECT INTEGRATION METHOD

Exercise 2.3

Determine the deflection δ_B and slope θ_B for beam below, in terms of m , L & EI . Assume $EI = \text{constant}$.



3. THE MACAULAY'S METHOD

3.1 Introduction

NOTE

1. The direct integration method, as outlined in chapter 2, is quite efficient for beams made up of one region only i.e. beams which only require one bending moment equation for the entire length. (See examples 2.1 and 2.2 above)
2. However the direct integration method becomes tedious and too involving if the BMD is piece-wise continuous. In each region there will be two constants of integration to be solved for! (See example 2.3). This can lead to a large set of equations.
3. This apparent difficulty was overcome by Macaulay, as discussed in Chapter 3. Macaulay's method is a modification of the direct integration method, based on the characteristics of singularity functions, which enables us to overcome the above difficulty.

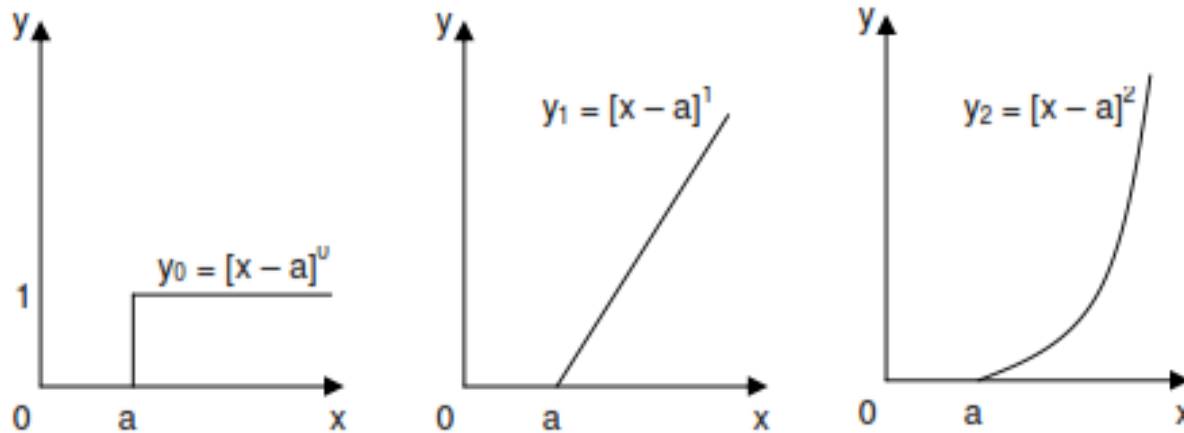
3.2 Singularity Functions

Consider the mathematical functions, called singularity functions, shown below. The singularity functions are expressed by use of square brackets, thus:

$$y_n = [x - a]^n \quad \text{for } n = 0, 1, 2, 3, \dots$$

3. THE MACAULAY'S METHOD

3.2 Singularity Functions



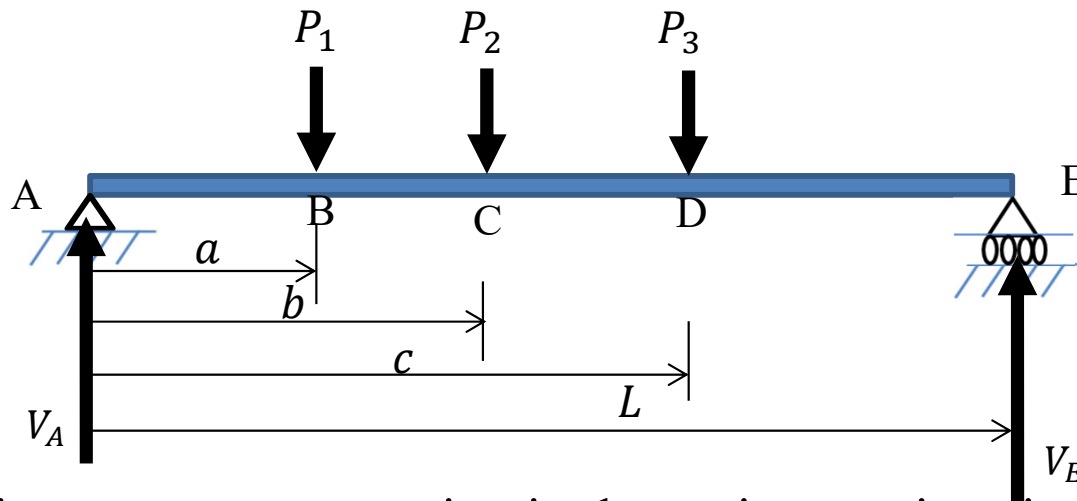
Singularity functions have three characteristics: -

1. $y_n = [x - a]^n = 0$ for all $x \leq a$; The function is non-existent for all values of x equal to or less than **a**
2. $y_n = [x - a]^n = 0$ for all $x > a$; The function takes its actual mathematical value for all values of x greater than a . [Once the function starts, it continues to the “end” of x – axis]
3. $\int [x - a]^n . dx = \frac{[x - a]^{n+1}}{n+1} + C$ for all $n \geq 0$; The square bracket is treated as the variable during integration.

3. THE MACAULAY'S METHOD

3.3 Macaulay's Moment Expression

Consider a simply supported beam subjected to point loads as shown below.



The bending moment expression in the various regions is conventionally expressed as: -

Region AB: $0 \leq x \leq a$: $M = V_A \cdot x$

Region BC: $a \leq x \leq b$: $M = V_A \cdot x - P_1(x - a)$

Region CD: $b \leq x \leq c$: $M = V_A \cdot x - P_1(x - a) - P_2(x - b)$

Region DE: $c \leq x \leq L$: $M = V_A \cdot x - P_1(x - a) - P_2(x - b) - P_3(x - c)$

According to Macaulay, a single bending moment expression for the whole beam may be written, using square brackets, as:

$$M = V_A \cdot x - P_1[x - a] - P_2[x - b] - P_3[x - c]$$

3. THE MACAULAY'S METHOD

Note:

1. In the above expression, all the square brackets are to power 1. Thus the bending moment diagram is piece-wise linear
2. The expression is valid over the whole beam, because characteristics number 1 and 2 of singularity functions apply.

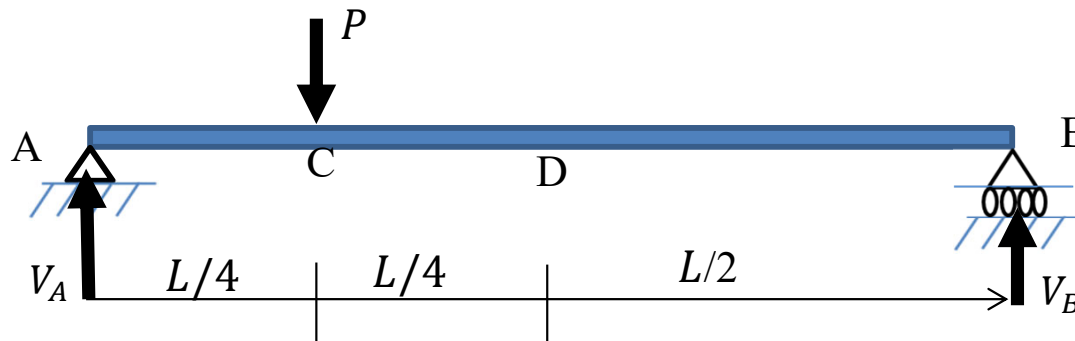
3.4 Solution Procedure

1. Determine support reaction components to obtain a free body diagram.
2. Use square brackets to express the bending moment in the last region of the beam.
3. Integrate twice, observing characteristic number 3.
4. Apply boundary conditions, observing characteristic no. 1 and 2; solve for C
5. Obtain mathematical expressions for the deflection and slope profiles.
6. Use these expressions to evaluate the required displacements.
7. Sketch the D-shape.

3. THE MACAULAY'S METHOD

Worked Example 3.1

Determine the deflection at under the point load and at mid-span and slopes θ_A & θ_B at the supports of the simply supported beam below, in terms of P , L and EI . Assume $EI = \text{constant}$.



Support Reactions

$$\sum F_v = 0 \Rightarrow V_A + V_B = P$$

And

$$\sum M \text{ at } A = 0$$

$$\Rightarrow V_B \cdot L = LP/4$$

$$\Rightarrow V_B = P/4$$

$$\text{and } V_A = 3P/4$$

Region CB

$$M_x = \frac{3Px}{4} - P\left[x - \frac{L}{4}\right]$$

$$EI \frac{d^2y}{dx^2} = -\frac{3Px}{4} + P\left[x - \frac{L}{4}\right]$$

$$EI \frac{dy}{dx} = -\frac{3Px^2}{8} + \frac{P\left[x - \frac{L}{4}\right]^2}{2} + C_1$$

$$EI y = -\frac{3Px^3}{24} + \frac{P\left[x - \frac{L}{4}\right]^3}{6} + C_1x + C_2$$

Apply Boundary conditions,

$$1. \text{ At } x = 0, y = 0$$

$$\Rightarrow C_2 = 0$$

$$2. \text{ At } x = L, y = 0$$

$$0 = -\frac{3PL^3}{24} + \frac{P\left[L - \frac{L}{4}\right]^3}{6} + LC_1 + 0$$

$$\Rightarrow C_1 = \frac{3PL^2}{24} - \frac{27PL^2}{384} = \frac{21PL^2}{384} = \frac{7PL^2}{128}$$

3. THE MACAULAY'S METHOD

Deflection and Slope profiles

$$EI \frac{dy}{dx} = -\frac{3Px^2}{8} + \frac{P\left[x - \frac{L}{4}\right]^2}{2} + \frac{7PL^2}{128} \Rightarrow \text{slope profile}$$

$$EI y = -\frac{3Px^3}{24} + \frac{P\left[x - \frac{L}{4}\right]^3}{6} + \frac{7PL^2x}{128} \Rightarrow \text{deflection profile}$$

Slope at A

$$, x = 0 \therefore \text{Slope } \theta_A = \frac{7PL^2}{128EI}$$

Deflection at C

$$, x = L/4$$

$$\therefore \text{Deflection } \delta_C = -\frac{3PL^3}{1536} + 0 + \frac{7PL^3}{512} = \frac{3PL^3}{256EI}$$

Slope at B

$$, x = L \therefore \text{Slope } \theta_B = -\frac{3PL^2}{8} + \frac{P\left[L - \frac{L}{4}\right]^2}{2} + \frac{7PL^2}{128} = -\frac{5PL^2}{128EI}$$

Deflection at Mid-span, D

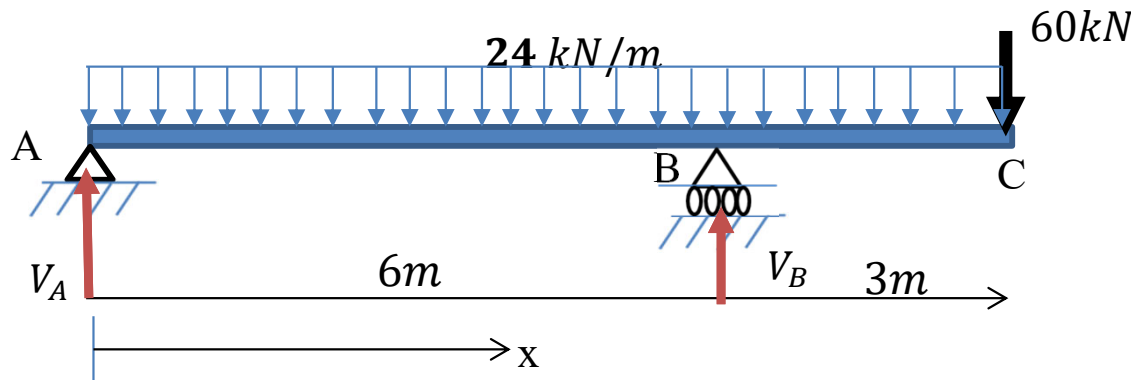
Where mid-span lies at, $x = L/2$

$$\therefore \text{Deflection } \delta_D = -\frac{3PL^3}{192} + \frac{PL^3}{384} + \frac{7PL^3}{256} = \frac{11PL^3}{768EI}$$

3. THE MACAULAY'S METHOD

Worked Example 3.2

Determine δ_C , θ_C and δ_{max} within span AB stating where it occurs. Assume $EI = \text{constant}$.



Solution

Support Reactions

$$\sum F_v = 0 \Rightarrow V_A + V_B = (24 \times 9) + 60$$

$$\Rightarrow V_A + V_B = 276 \text{ kN}$$

$$\sum M \text{ at } A = 0$$

$$\Rightarrow 6V_B = (60 \times 9) + (24 \times 9 \times 4.5)$$

$$\Rightarrow 6V_B = 1512$$

$$\Rightarrow V_B = \mathbf{252 \text{ kN}}$$

$$\text{and } V_A = 276 - 252 = \mathbf{24 \text{ kN}}$$

Region BC

$$M_x = 24x - \frac{24x^2}{2} + 252[x - 6]$$

$$EI \frac{d^2y}{dx^2} = -24x + \frac{24x^2}{2} - 252[x - 6]$$

$$EI \frac{dy}{dx} = -\frac{24x^2}{2} + \frac{24x^3}{6} - \frac{252[x - 6]^2}{2} + C_1$$

$$EI y = -\frac{24x^3}{6} + \frac{24x^4}{24} - \frac{252[x - 6]^3}{6} + C_1x + C_2$$

3. THE MACAULAY'S METHOD

Apply Boundary conditions,

1. At $x = 0, y = 0 \Rightarrow C_2 = 0$

2. At $x = 6m, y = 0$

$$0 = -\frac{24 \times 6^3}{6} + \frac{24 \times 6^4}{24} - \frac{252[6-6]^3}{6} + 6C_1 + 0$$

$$6C_1 = \frac{24 \times 6^3}{6} - \frac{24 \times 6^4}{24} + 0 \quad \text{or } C_1 = -72$$

Deflection and Slope profiles

$$EI \frac{dy}{dx} = -\frac{24x^2}{2} + \frac{24x^3}{6} - \frac{252[x-6]^2}{2} - 72 \Rightarrow \text{slope profile}$$

$$EI y = -\frac{24x^3}{6} + \frac{24x^4}{24} - \frac{252[x-6]^3}{6} - 72x \Rightarrow \text{deflection profile}$$

Slope at C

$$, x = 9 \therefore \text{Slope } \theta_C = \left(-\frac{24 \times 9^2}{2} + \frac{24 \times 9^3}{6} - \frac{252[9-6]^2}{2} - 72 \right) / EI = 738 / EI$$

Deflection at C

$$, x = 9$$

$$\therefore \text{Deflection } \delta_C = \left(-\frac{24 \times 9^3}{6} + \frac{24 \times 9^4}{24} - \frac{252[9-6]^3}{6} - 72 \times 9 \right) / EI = 1863 / EI$$

3. THE MACAULAY'S METHOD

δ_{max} within span AB

Occurs when slope = 0 or $EI \frac{dy}{dx} = 0$

$$\text{Or } 0 = -\frac{24x^2}{2} + \frac{24x^3}{6} - \frac{252[x-6]^2}{2} - 72$$

But within span AB, $x \leq 6 \therefore [x-6]^2 = 0$

$$\text{Or } 0 = -12x^2 + 4x^3 - 0 - 72$$

$$\text{Or } 4x^3 - 12x^2 - 72 = 0$$

Solving the cubic equation gives $x = 4.081m$

Substituting for $x = 4.081m$ in the deflection profile we get,

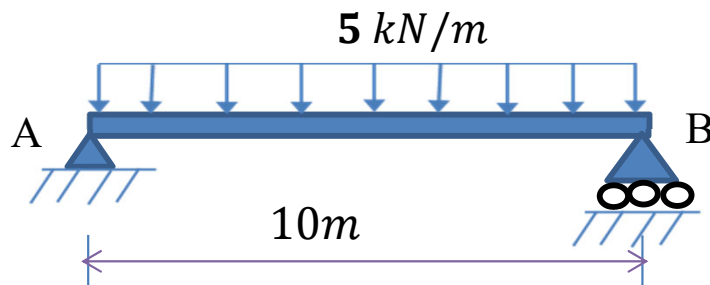
$$EI \delta_{max} = -\frac{24 \times 4.081^3}{6} + \frac{24 \times 4.081^4}{24} - 0 - 72 \times 4.081$$

$$\delta_{max} = -288.33/EI$$

3. THE MACAULAY'S METHOD

Exercise 3.1

Determine the deflection at mid-span and slopes θ_A & θ_B at the supports of the simply supported beam below using the Macaulay's method. Assume $EI =$ constant.



$$\delta_{Mid\ Span} = 25 \times 10^4 / 384 EI$$

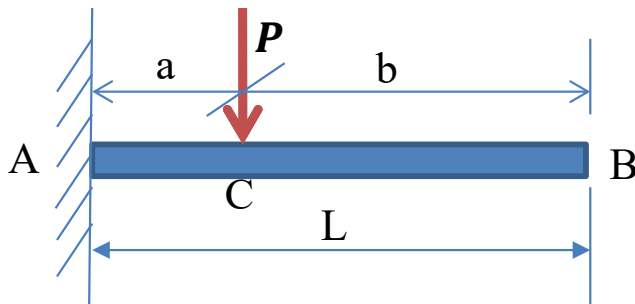
$$= 651.042 / EI$$

$$\theta_A = 5 \times 10^3 / 24 EI = 208.333 / EI$$

$$\theta_B = -5 \times 10^3 / 24 EI = -208.333 / EI$$

Exercise 3.2

Using the Macaulay's method, evaluate the deflection at C & B and slopes θ_C & θ_B for beam below, in terms of P , L , a , b & EI . Assume $EI =$ constant.



$$\delta_C = Pa^3 / 3EI$$

$$\theta_C = Pa^2 / 2EI$$

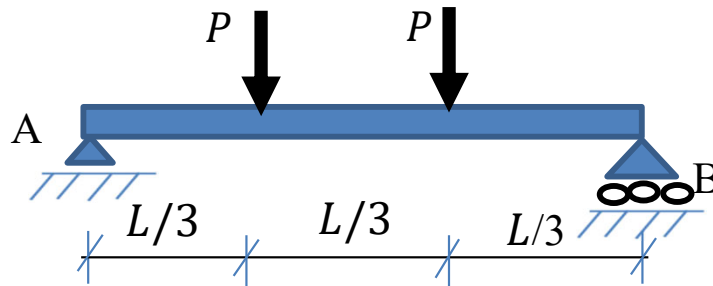
$$\delta_B = Pa^2 \{3(a+b) - a\} / 6EI$$

$$\theta_B = Pa^2 / 2EI$$

3. THE MACAULAY'S METHOD

Exercise 3.3

Using the Macaulay's method, evaluate the maximum deflection δ_{max} and slopes θ_A of the simply supported beam below. Assume $EI = \text{constant}$.

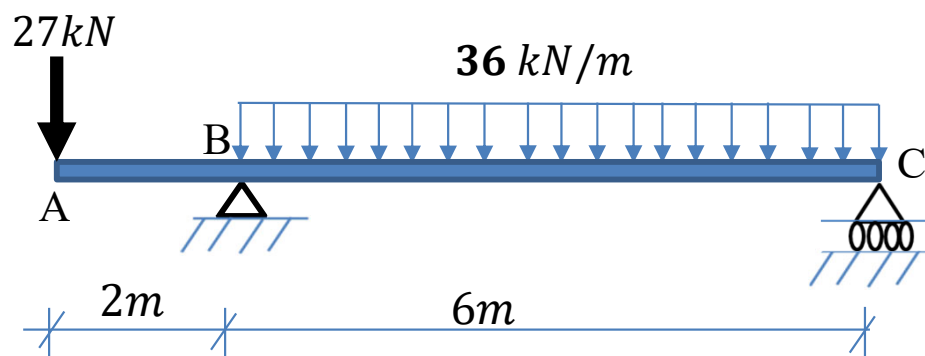


$$\delta_{max} = 23PL^3/648EI \text{ at } x = L/2$$

$$\theta_A = PL^2/9EI$$

Exercise 3.4

Using the Macaulay's method, evaluate the deflection δ_A & δ_{max} within span BC and slopes θ_A , θ_B & θ_C for beam below, in terms of P , L , a , b & EI . Assume $EI = \text{constant}$.



$$\delta_A = -360/EI$$

$$\delta_{max} \text{ within } BC = 486.668/EI \text{ at } x = 5.1m$$

$$\theta_A = 162/EI$$

$$\theta_B = 216/EI$$

$$\theta_C = -270/EI$$

4. THE MOMENT -AREA METHOD

4.1 Introduction

The Moment-Area method was developed by Charles Ezra Greene (1897) based on Mohr's theorems (1870), which relate the geometry of the deflected shape to the curvature of the beam.

MOHR'S 1ST THEOREM

Using the expression

$$\frac{d^2y}{dx^2} = -M/EI$$

and noting that $\theta = dy/dx$ we can express: $-d\theta/dx = M/EI$

or $-d\theta = (M/EI) dx$ Integrating both sides gives:

$$-\int_A^B d\theta = \int_A^B (M/EI) dx \quad \text{Or } -(\theta_B - \theta_A) = \int_A^B (M/EI) dx$$

$$\text{Or } \theta_A - \theta_B = \int_A^B (M/EI) dx \text{----- Eqn 1}$$

Mohr's 1st theorem states that the change in slope due to flexure between any two points, say A and B, is equal to the area under the M/EI diagram between the two points.

4. THE MOMENT -AREA METHOD

MOHR'S 2ND THEOREM

The expression

$\frac{d^2y}{dx^2} = -M/EI$ may be multiplied by x on both sides and then integrated to obtain:

$$-\int_A^B x \cdot \frac{d^2y}{dx^2} \cdot dx = \int_A^B x \cdot (M/EI) dx \dots\dots\dots \text{Eqn.(i)}$$

The R.H. side of the above expression represents the 1st moment of area of the M/EI diagram between A and B, taken about A. The L.H. side may be treated as the product of the function v and the differential du .

In calculus, $\int v du = uv - \int u dv$, and therefore the L.H. side may be written as:

$$-\left\{ x \cdot \frac{dy}{dx} \right\}_A^B - \int_A^B \frac{dy}{dx} \cdot dx = -\{ (X_B \theta_B - X_A \theta_A) - (Y_B - Y_A) \} = -\{ X_B \theta_B - X_A \theta_A - Y_B + Y_A \}$$

$$\text{Or } -\left\{ x \cdot \frac{dy}{dx} \right\}_A^B - \int_A^B \frac{dy}{dx} \cdot dx = (Y_B - X_B \theta_B) - (Y_A - X_A \theta_A) = \int_A^B x \cdot (M/EI) dx$$

If we set our reference axis to originate from A, then $Y_A = X_A = 0$, and

$$-\left\{ x \cdot \frac{dy}{dx} \right\}_A^B - \int_A^B \frac{dy}{dx} \cdot dx = (Y_B - X_B \theta_B)$$

4. THE MOMENT -AREA METHOD

Hence eqn. (i) becomes: $(Y_B - X_B \theta_B) = \int_A^B x \cdot (M/EI) dx \dots \dots \text{Eqn. (ii)}$

The quantity $(Y_B - X_B \theta_B)$ is the vertical deviation of the point A from a tangent drawn at the point B, and is denoted by $\Delta_{A/B}$. Eqn (ii) is therefore written as

$$\Delta_{A/B} = \int_A^B x \cdot (M/EI) dx \text{ ----- Eqn 2}$$

Mohr's 2nd theorem states that the vertical deviation of point A from the tangent drawn at point B is equal to the 1st moment of area of the M/EI diagram between A and B, taken about A. [$\Delta_{A/B}$ is measured positive upwards from the tangent drawn at B].

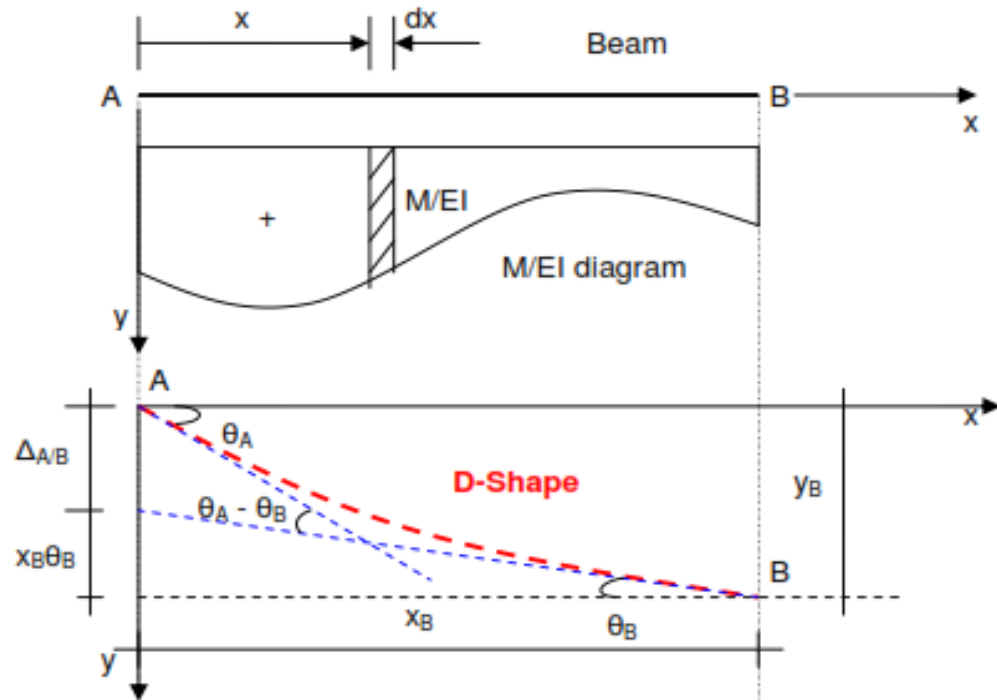
4.2. Interpretation of Mohr's Theorems

Mohr's theorems may be interpreted diagrammatically as shown below.

4. THE MOMENT -AREA METHOD

4.2. Interpretation of Mohr's Theorems

Mohr's theorems may be interpreted diagrammatically as shown below.



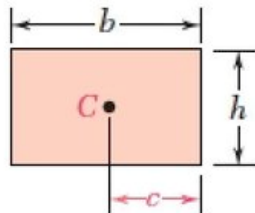
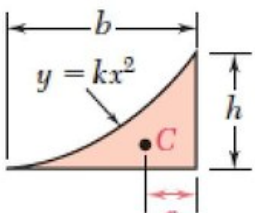
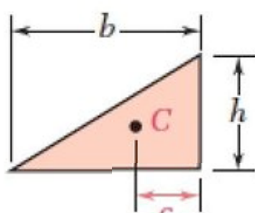
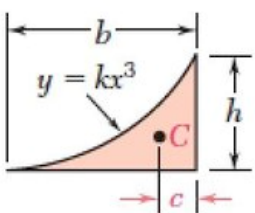
4.3 Solution Procedure

The solution procedure may be outlined as follows: -

1. Analyze the beam to obtain the M/EI diagram.
2. Sketch the D-shape.
3. Use 1st theorem to calculate slopes. [One of the slopes must be known]
4. Use 2nd theorem to calculate deflections. [Define the point of reference and its tangent carefully]

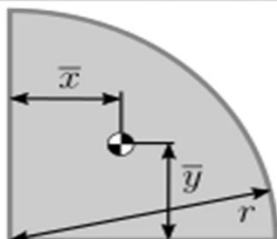
4. THE MOMENT -AREA METHOD

4.4 Centroid of Some Shapes

Shape		Area	c	Shape		Area	c
Rectangle		bh	$\frac{b}{2}$	Parabolic spandrel		$\frac{bh}{3}$	$\frac{b}{4}$
Triangle		$\frac{bh}{2}$	$\frac{b}{3}$	Cubic spandrel		$\frac{bh}{4}$	$\frac{b}{5}$

Quarter Circular area

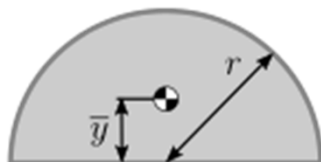
$$\bar{x} = \bar{y} = \frac{4r}{3\pi}$$



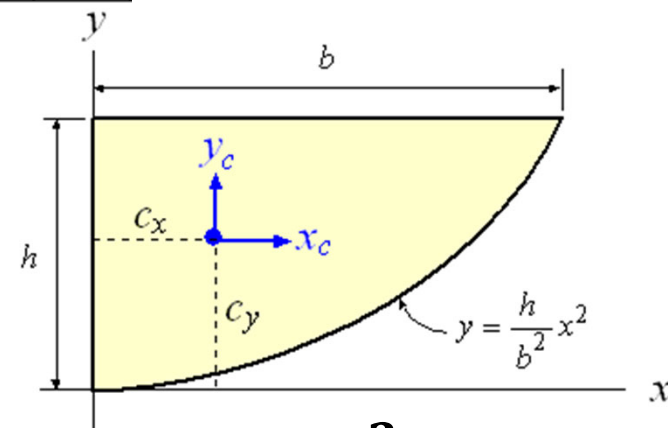
Semi-Circular area

$$\bar{x} = 0$$

$$\bar{y} = \frac{4r}{3\pi}$$



Quarter Parabola



$$C_x = \frac{3}{8}b$$

$$C_y = \frac{3}{5}h$$

$$Area = \frac{2}{3}bh$$

4. THE MOMENT -AREA METHOD

Worked Example 4.1

Determine the deflection δ_B and slope θ_B at the free end B of the cantilever beam below, in terms of P, L and EI. Assume EI = constant.

Solution

Using 1st theorem, $\theta_A - \theta_B = \int_A^B (M/EI) dx$

Or $\theta_A - \theta_B = \frac{1}{2} \times L \times \frac{-PL}{EI}$ but $\theta_A = 0$

$$\therefore -\theta_B = \frac{-PL^2}{2EI} \quad \text{Or} \quad \therefore \theta_B = \frac{PL^2}{2EI}$$

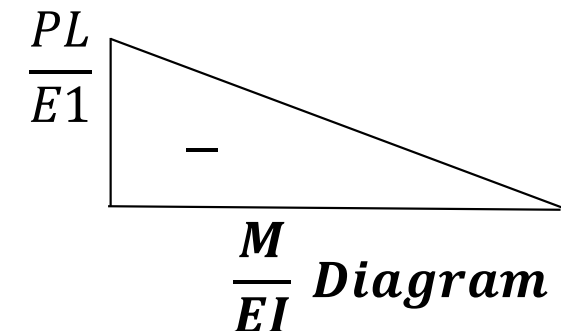
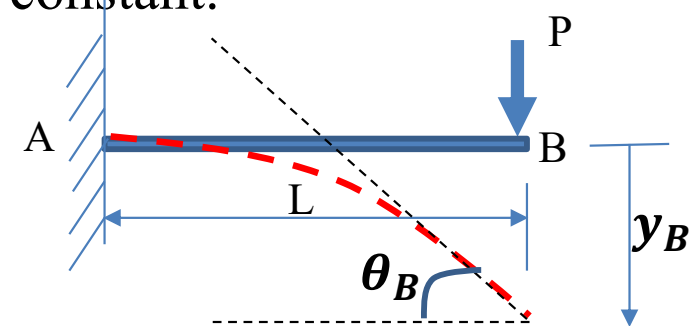
Using 2nd theorem;

$$(Y_B - X_B \theta_B) - (Y_A - X_A \theta_A) = \int_A^B x \cdot (M/EI) dx$$

$$-\Delta_{B/A} = \left(\frac{1}{2}\right)(L) \left(-\frac{PL}{EI}\right) \left(\frac{2}{3}\right)(L)$$

$$\Delta_{B/A} = PL^3/3EI$$

By geometry $\delta_B = \Delta_{B/A} = PL^3/3EI$



4. THE MOMENT -AREA METHOD

Worked Example 4.2

Determine the deflection δ_B and slope θ_B at the free end B of the cantilever beam below, in terms of P, L and EI. Assume EI = Varies as shown.

Solution

Using 1st theorem, $\theta_A - \theta_B = \int_A^B (M/EI) dx$

$$\text{Or } \theta_A - \theta_B = \left(\frac{1}{2} \times \frac{L}{2} \times \frac{-PL}{2EI} \right) + \left(\frac{1}{2} \times \frac{L}{2} \times \frac{-PL}{4EI} \right) + \left(\frac{L}{2} \times \frac{-PL}{4EI} \right) = \frac{-5PL^2}{16EI} \quad \text{but } \theta_A = 0$$

$$\therefore -\theta_B = \frac{-5PL^2}{16EI} \quad \text{Or } \therefore \theta_B = \frac{5PL^2}{16EI}$$

Using 2nd theorem;

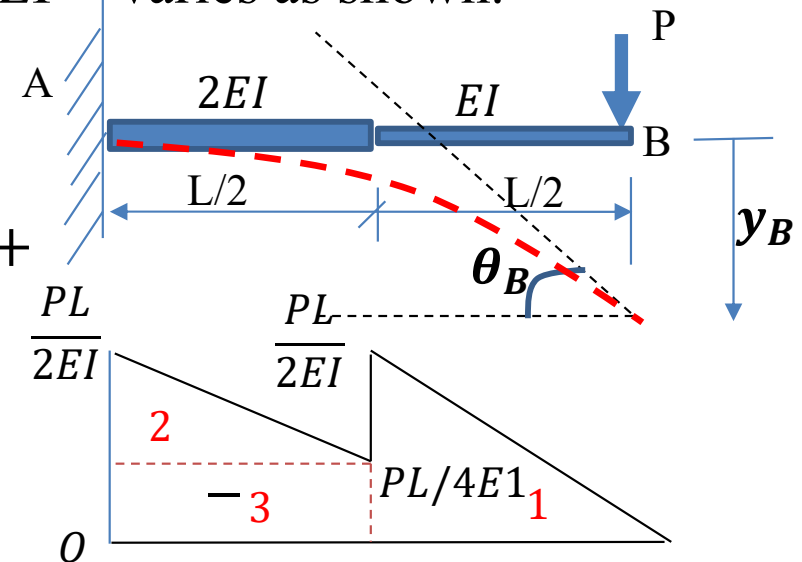
$$(Y_B - X_B \theta_B) - (Y_A - X_A \theta_A) =$$

$$\int_A^B x \cdot (M/EI) dx$$

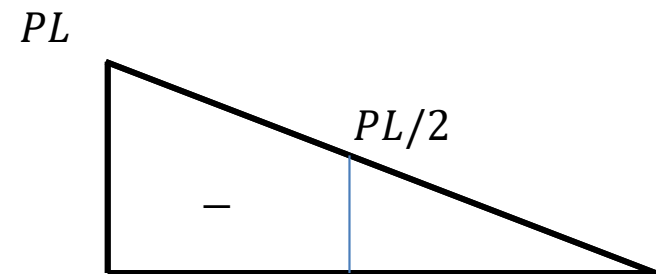
$$-\Delta_{B/A} = \left(\frac{1}{2} \times \frac{L}{2} \times \frac{-PL}{2EI} \times \frac{2}{3} \times \frac{L}{2} \right) + \left(\frac{1}{2} \times \frac{L}{2} \times \frac{-PL}{4EI} \times \frac{5L}{6} \right) + \left(\frac{L}{2} \times \frac{-PL}{4EI} \times \frac{3L}{4} \right) = \frac{-18PL^3}{96EI}$$

$$\Delta_{B/A} = 3PL^3/16EI$$

By geometry $\delta_B = \Delta_{B/A} = 3PL^3/16EI$



$\frac{M}{EI}$ Diagram



4. THE MOMENT -AREA METHOD

Worked Example 4.3

Determine the deflection δ_B and slope θ_B at the free end B of the cantilever beam below, in terms of w , L and EI . Assume $EI = \text{constant}$.

Solution

Using 1st theorem, $\theta_A - \theta_B = \int_A^B (M/EI) dx$

$$\text{Or } \theta_A - \theta_B = \left(\frac{1}{3} \times L \times \frac{-wL^2}{2EI} \right) = \frac{-wL^3}{6EI}$$

$$\text{but } \theta_A = 0 \therefore -\theta_B = \frac{-wL^3}{6EI} \text{ Or } \therefore \theta_B = \frac{wL^3}{6EI}$$

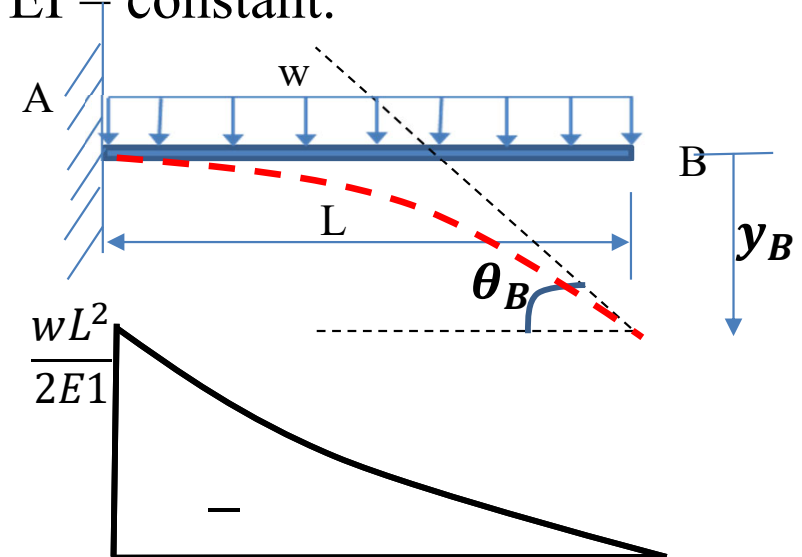
Using 2nd theorem;

$$(Y_B - X_B \theta_B) - (Y_A - X_A \theta_A) = \int_A^B x \cdot (M/EI) dx$$

$$-\Delta_{B/A} = \left(\frac{1}{3} \times L \times \frac{-wL^2}{2EI} \right) \times \frac{3}{4} L = \frac{-wL^4}{8EI}$$

$$\Delta_{B/A} = wL^4/8EI$$

$$\text{By geometry } \delta_B = \Delta_{B/A} = wL^4/8EI$$



$\frac{M}{EI}$ Diagram

4. THE MOMENT -AREA METHOD

Worked Example 4.4

Determine the deflection δ_C and slope θ_A for the beam below, in terms of w , L and EI . Assume $EI = \text{constant}$.

Solution

Using 1st theorem, $\theta_A - \theta_B =$

$$\int_A^B (M/EI) dx$$

$$\text{Or } \theta_A - \theta_C = \left(\frac{2}{3} \times \frac{L}{2} \times \frac{wL^2}{8EI} \right) = \frac{wL^3}{24EI}$$

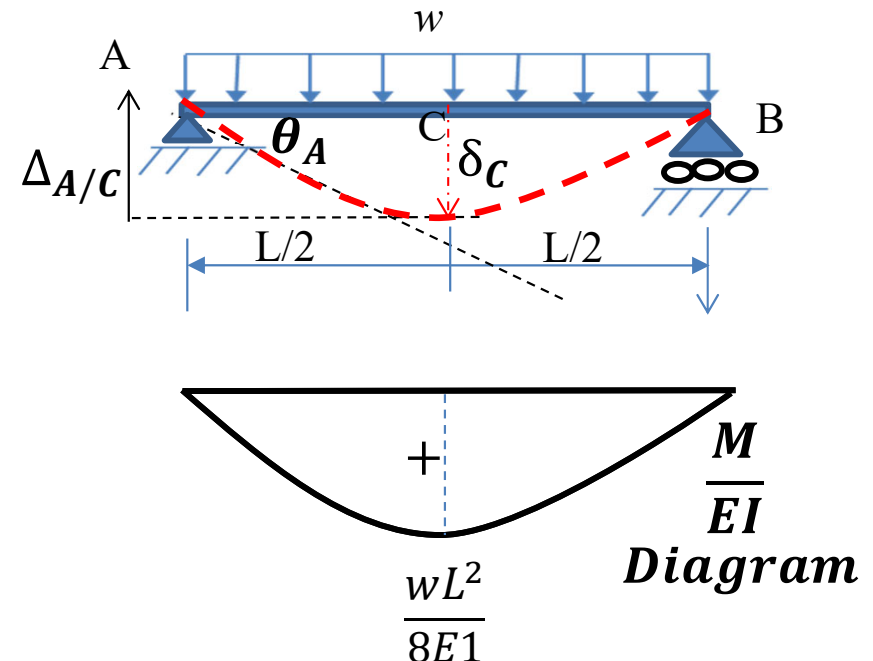
$$\text{but } \theta_C = 0 \therefore \theta_A = \frac{wL^3}{24EI}$$

Using 2nd theorem;

$$\Delta_{A/C} = \left(\frac{2}{3} \times \frac{L}{2} \times \frac{wL^2}{8EI} \right) \times \left(\frac{5}{8} \times \frac{L}{2} \right) = \frac{5wL^4}{384EI}$$

$$\Delta_{A/C} = 5wL^4/384EI$$

$$\text{By geometry } \delta_C = \Delta_{A/C} = 5wL^4/384EI$$

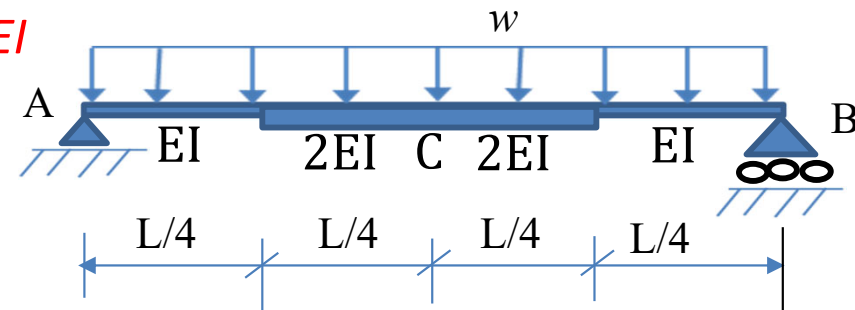


4. THE MOMENT -AREA METHOD

Exercise 4.1

Using the moment-area method, determine the deflection δ_C and slope θ_A for the beam below, in terms of w , L and EI . Assume EI = Varies as shown

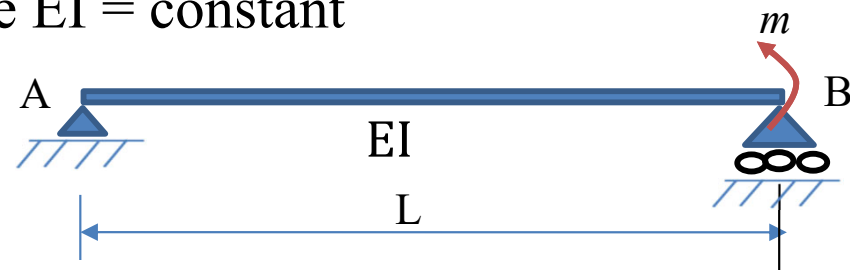
$$\delta_C = 31wL^4 / 4096EI, \theta_A = 7wL^3 / 256EI$$



Exercise 4.2

Using the moment-area method, determine the slopes θ_A and θ_B for the beam below, in terms of m , L and EI . Assume EI = constant

$$\theta_A = -mL / 6EI, \theta_B = mL / 3EI$$

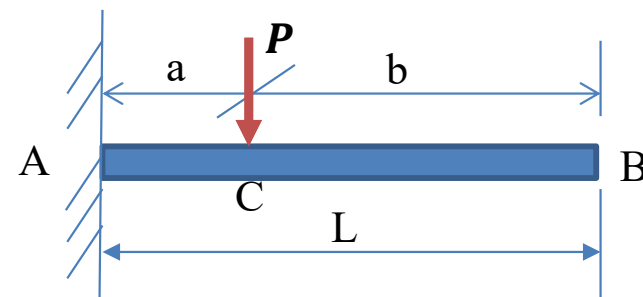


Exercise 4.3

Using the moment area method, evaluate δ_C , δ_B , θ_C & θ_B for beam below, in terms of P , L , a , b & EI . Assume EI = constant.

$$\delta_C = Pa^3 / 3EI, \theta_C = Pa^2 / 2EI$$

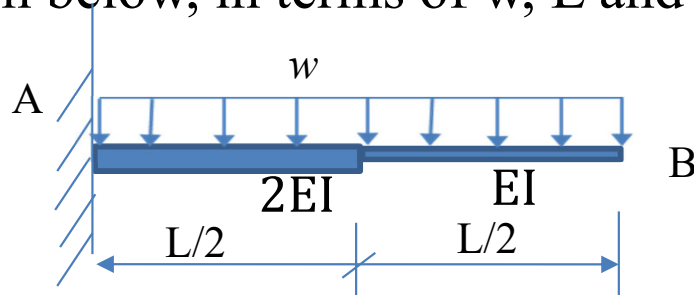
$$\delta_B = Pa^2 \{ 3(a+b) - a \} / 6EI, \theta_B = Pa^2 / 2EI$$



4. THE MOMENT -AREA METHOD

Exercise 4.4

Using the moment-area method, evaluate the deflection δ_B and slope θ_B for the beam below, in terms of w , L and EI . Assume EI = Varies as shown

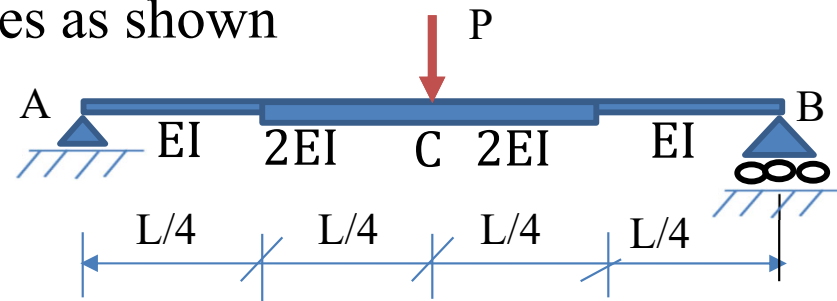


$$\delta_B = 17wL^4 / 256EI, \quad \theta_B = 3wL^3 / 32EI$$

Exercise 4.5

Using the moment-area method, evaluate θ_A and δ_C for the beam below, in terms of P , L and EI . Assume EI = Varies as shown

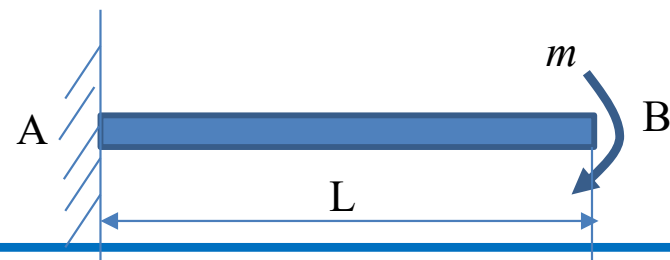
$$\delta_C = 3PL^3 / 256EI, \quad \theta_A = 5PL^2 / 128EI$$



Exercise 4.6

Using the moment area method, evaluate δ_B & θ_B for beam below, in terms of P , L , a , b & EI . Assume EI = constant.

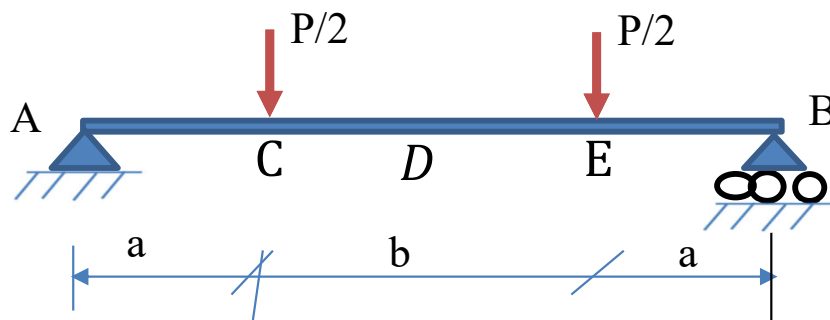
$$\delta_B = mL^2 / 2EI, \quad \theta_B = mL / EI$$



4. THE MOMENT -AREA METHOD

Exercise 4.7

Using the moment-area method ,evaluate θ_A , θ_C , δ_C **and** δ_D (at mid span) for the beam below, in terms of a, b, P and EI. Assume EI = constant



$$\theta_A = Pa(a + b)/4EI$$

$$\theta_C = Pab/4EI$$

$$\delta_C = Pa^3/6EI ,$$

$$\delta_D = Pa(8a^2 + 12ab + 3b^2)/48EI$$

5. THE CONJUGATE BEAM METHOD

5.1: Introduction

The Conjugate Beam Method was developed by Muller-Breslau (1895) based on Mohr's Analogy (1870)

MOHR'S ANALOGY (1870)

- Mohr noticed that there exists some analogy between load-shear-moment expressions in and the curvature-slope-deflection expressions
- Mohr concluded that the load must be analogous to the curvature, the shear must be analogous to the slope, and the moment must be analogous to the deflection.

$$\begin{aligned}
 1. \quad \text{Load: } \frac{d^2 M}{dx^2} &= -w & \Rightarrow & \quad \text{Curvature: } \frac{d^2 y}{dx^2} = -\frac{M}{EI} \\
 2. \quad \text{Shear: } \frac{dM}{dx} &= V & \Rightarrow & \quad \text{Slope: } \frac{dy}{dx} = \theta \\
 3. \quad \text{Moment : } M &= \int V \cdot dx & \Rightarrow & \quad \text{Deflection: } y = \int \theta \cdot dx
 \end{aligned}$$

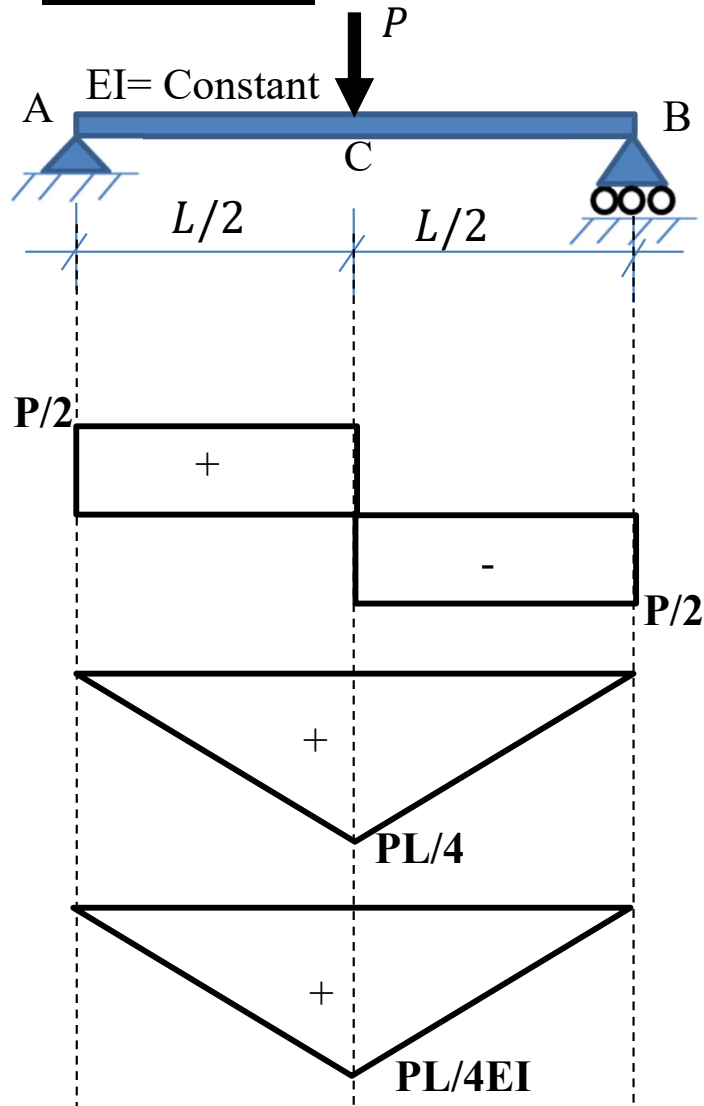
Worked Example 5.1

Consider the simply supported Real Beam shown below. The real beam is analyzed accordingly to obtain its BMD. The BMD is converted into the M/EI diagram, the curvature diagram.

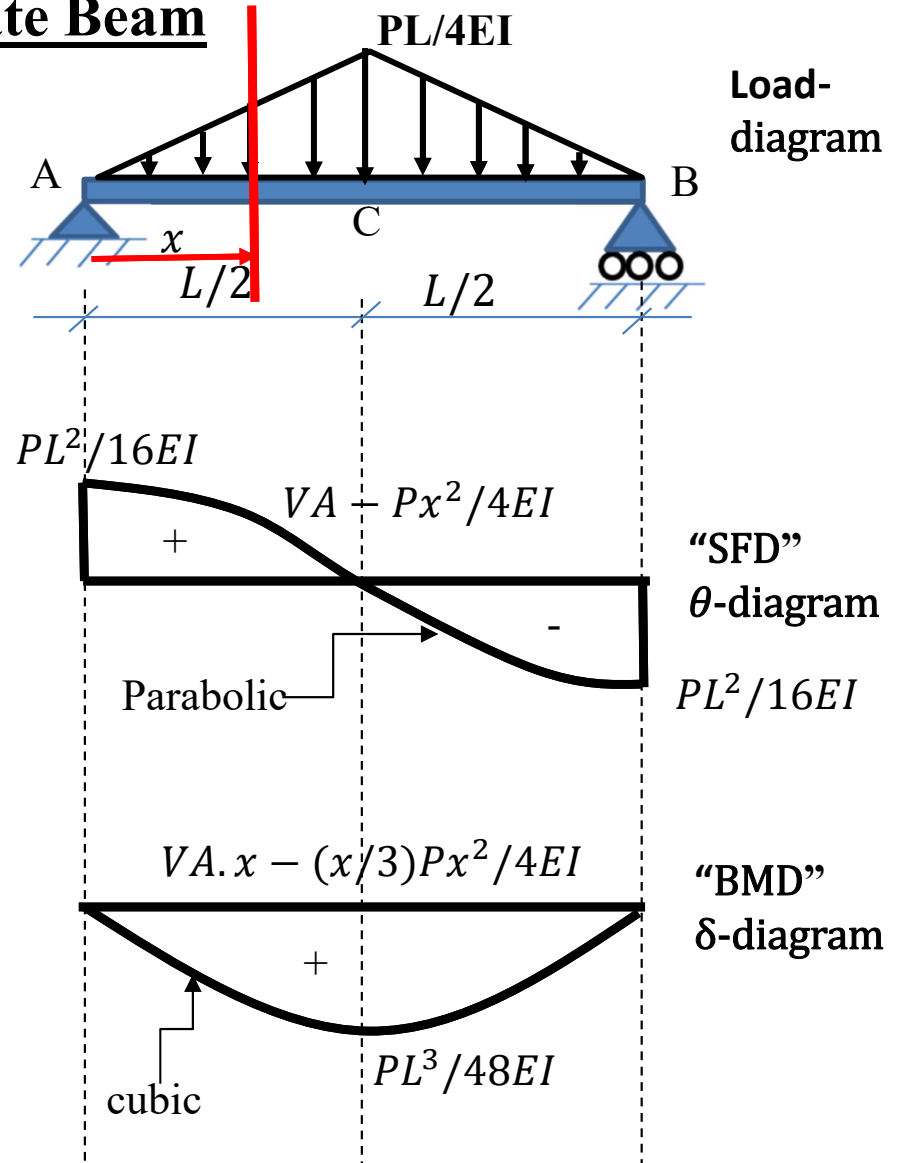
5. THE CONJUGATE BEAM METHOD

The M/EI diagram of the Real Beam is used as the “applied loading” on the Conjugate Beam. The conjugate beam is then analyzed accordingly to obtain its “SFD” and “BMD”.

Real Beam



Conjugate Beam



5. THE CONJUGATE BEAM METHOD

A careful inspection of these diagrams leads to the following conclusions: -

1. The shear force in the Conjugate Beam at any point is equal to the slope of the Real Beam at that point. Therefore, the “SFD” of the Conjugate Beam is actually the θ -diagram of the Real Beam. Thus $\theta_A = PL^2/16EI$, $\theta_B = -PL^2/16EI$, and $\theta_C = 0$
2. The bending moment in the Conjugate Beam at any point is equal to the deflection of the Real Beam at that point. Therefore, the “BMD” of the Conjugate Beam is actually the δ -diagram of the Real Beam. $\delta_A = \delta_B = 0$ and $\delta_C = PL^3/48EI$

Worked Example 5.2

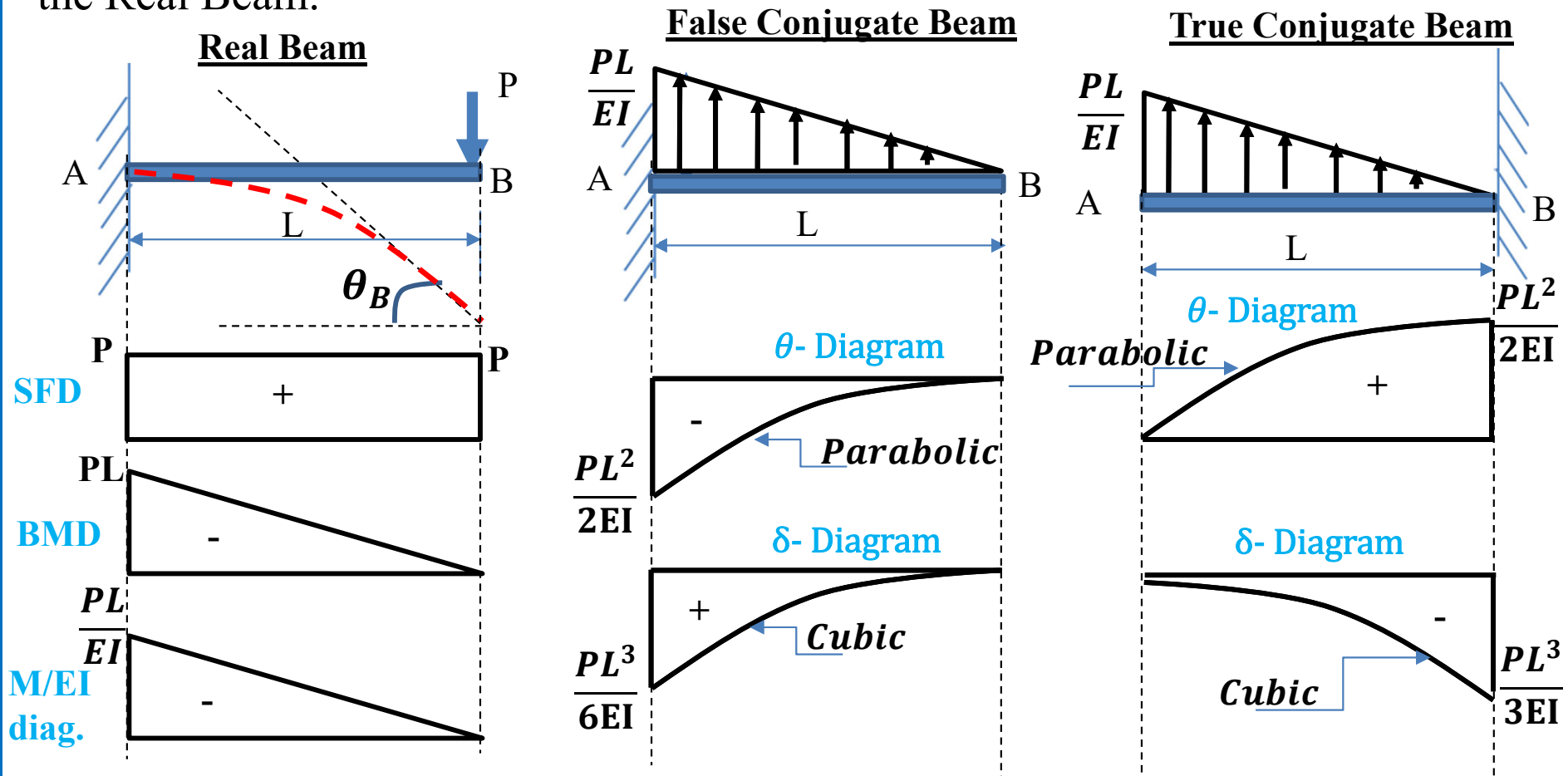
Consider the cantilever **Real Beam** shown below. The real beam is analysed to obtain its BMD. The BMD is converted into the M/EI diagram, the curvature diagram.

Consider the “**False Conjugate Beam**” shown alongside the Real Beam. The M/EI diagram of the Real Beam is used as the “applied loading” on the “False Conjugate Beam”. The “False Conjugate Beam” is then analysed to obtain its “SFD” and “BMD”. Analysis of the “False Conjugate Beam” shows that

$\delta_B = 0$ and $\delta_A = PL^3/6EI$. This is obviously **NOT correct**.

5. THE CONJUGATE BEAM METHOD

Consider the “**True Conjugate Beam**” shown alongside the Real Beam. The M/EI diagram of the Real Beam is used as the “applied loading” on the “True Conjugate Beam”. The “True Conjugate Beam” is then analysed to obtain its “SFD” and “BMD”. Analysis of the “True Conjugate Beam” shows that $\delta_A = 0$ and $\delta_B = PL^3/3EI$. This is obviously CORRECT. The “True Conjugate Beam” is obtained by transposing the slope and deflection boundary conditions of the Real Beam.



5. THE CONJUGATE BEAM METHOD

Conclusion

A conjugate beam is a beam whose shear and moment boundary conditions are the transpose of the corresponding slope and deflection boundary conditions of the real beam.

Transpose of Some Boundary Conditions

<u>Real beam</u>		<u>Conjugate beam</u>
1. Pin or Roller support at end $\delta = 0$ $\theta \neq 0$	becomes →	Pin or Roller support at end $M = 0$ $V \neq 0$
2. Fixed end $\delta = 0$ $\theta = 0$	→	Free end $M = 0$ $V = 0$
3. Free end $\delta \neq 0$ $\theta \neq 0$	→	Fixed end $M \neq 0$ $V \neq 0$
4. Internal hinge $\delta \neq 0$ $\theta \neq 0$	→	Continuous support $M \neq 0$ $V \neq 0$
5. Continuous support $\delta = 0$ $\theta \neq 0$	→	Internal hinge $M = 0$ $V \neq 0$

5. THE CONJUGATE BEAM METHOD

Solution Procedure

1. Analyse the Real Beam to obtain the M/EI diagram.
2. Configure the Conjugate Beam and load it with the M/EI diagram.
3. Analyse the conjugate beam to obtain the required displacements.

Note

Step 3 of the above procedure may involve dealing with linear and/or parabolic “load ”diagrams. The analysis can be made easier by use areas and centroids of some standard functions.

5. THE CONJUGATE BEAM METHOD

Worked Example 5.3

Determine the deflection θ_B and slope δ_B at the free end B of the cantilever beam below, in terms of P, L and EI. Assume EI = Varies as shown.

Solution

The slope at B is equal to the shear force at B on the conjugate beam. Thus :

$$\theta_B = \left(\frac{1}{2} \times \frac{L}{2} \times \frac{PL}{2EI}\right) + \left(\frac{1}{2} \times \frac{L}{2} \times \frac{PL}{4EI}\right) + \left(\frac{L}{2} \times \frac{PL}{4EI}\right)$$

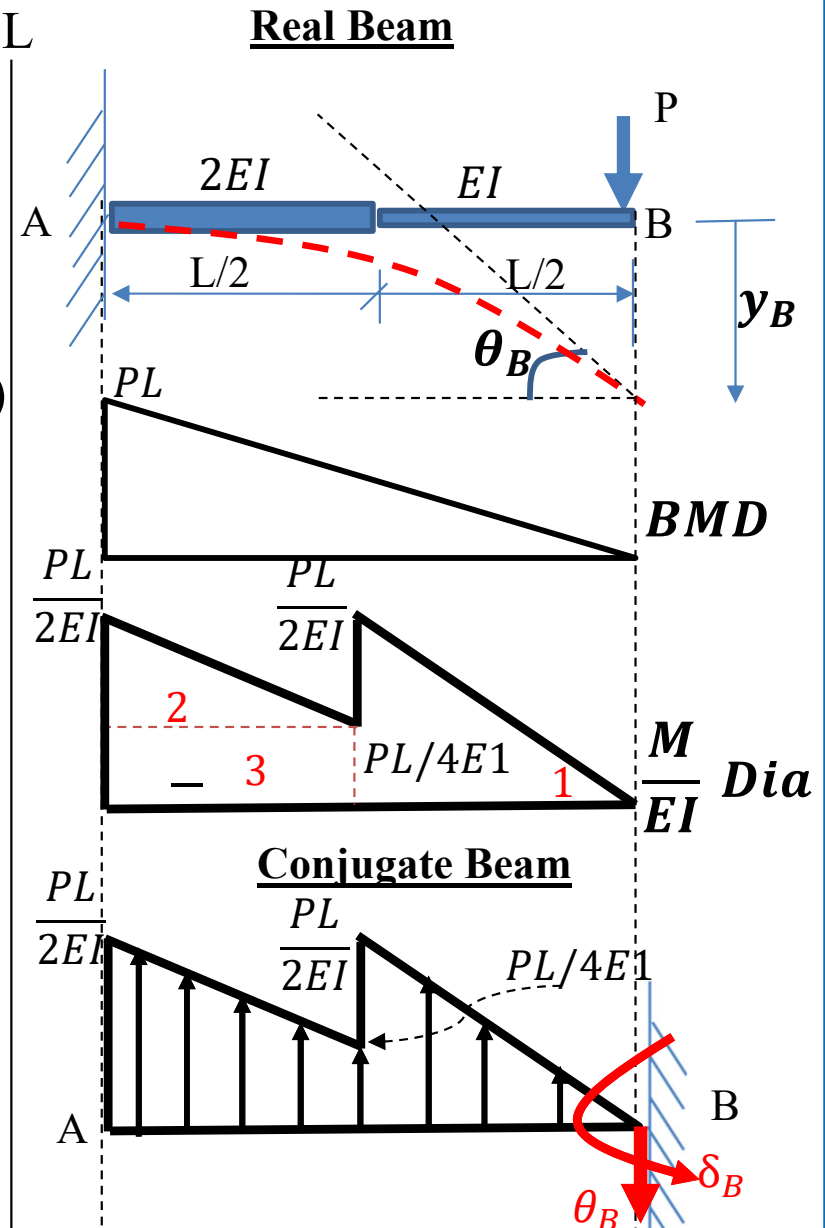
$$\theta_B = \frac{5PL^2}{16EI}$$

The deflection at B is equal to the moment at B on the conjugate beam. Thus:

$$\delta_B = \left(\frac{1}{2} \times \frac{L}{2} \times \frac{PL}{2EI} \times \frac{2}{3} \times \frac{L}{2}\right) +$$

$$\left(\frac{1}{2} \times \frac{L}{2} \times \frac{PL}{4EI} \times \frac{5L}{6}\right) + \left(\frac{L}{2} \times \frac{PL}{4EI} \times \frac{3L}{4}\right) = \frac{18PL^3}{96EI}$$

$$\delta_B = 3PL^3/16EI$$



5. THE CONJUGATE BEAM METHOD

Worked Example 5.4

Determine the deflection θ_B and slope δ_B at the free end B of the cantilever beam below, in terms of w , L and EI . Assume $EI = \text{Varies as shown}$.

Solution

The slope at B is equal to the shear force at B on the conjugate beam. Thus :

$$\theta_B = \left(\frac{1}{3} \times L \times \frac{wL^2}{4EI} \right) + \left(\frac{1}{3} \times \frac{L}{2} \times \frac{wL^2}{16EI} \right)$$

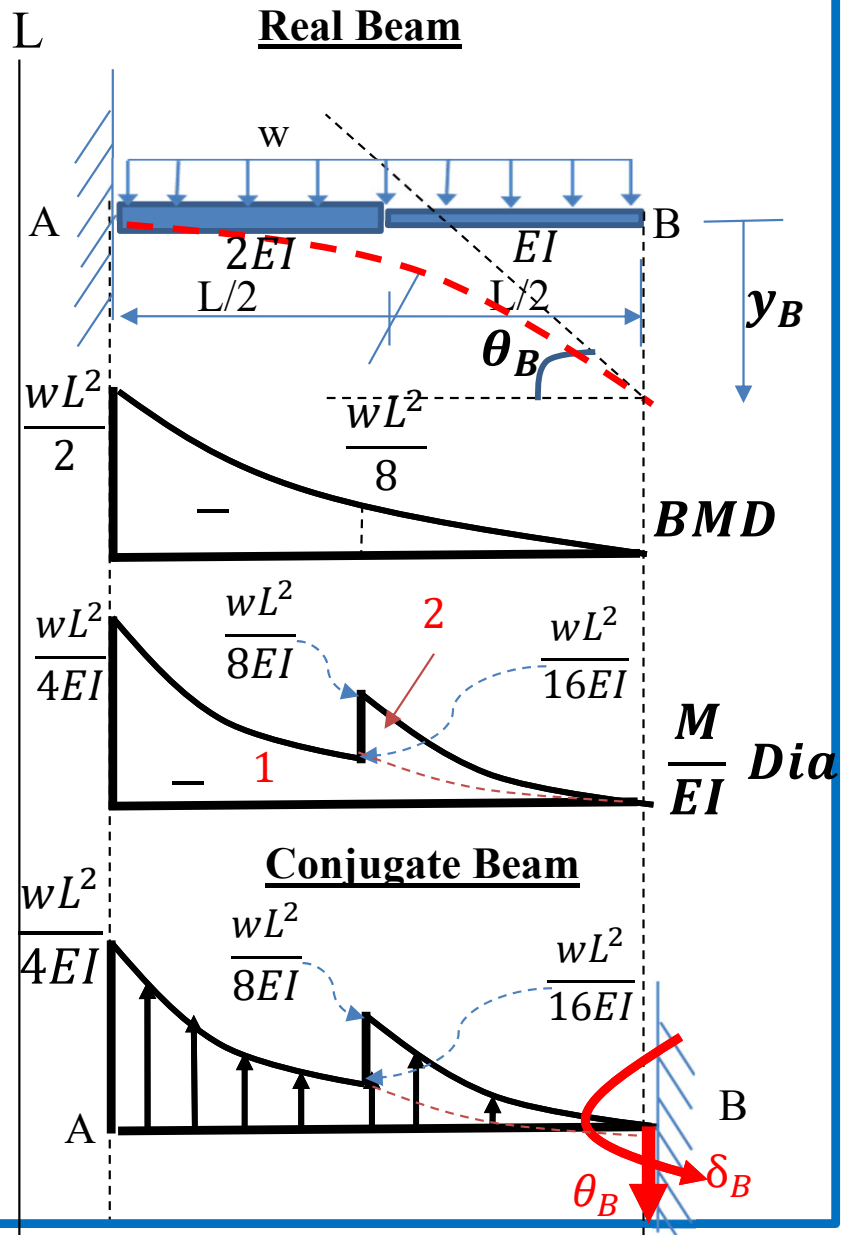
$$\theta_B = \frac{3wL^3}{32EI}$$

The deflection at B is equal to the moment at B on the conjugate beam. Thus:

$$\delta_B = \left(\frac{1}{3} \times L \times \frac{wL^2}{4EI} \times \frac{3}{4} \times L \right)$$

$$+ \left(\frac{1}{3} \times \frac{L}{2} \times \frac{wL^2}{16EI} \times \frac{3}{4} \times \frac{L}{2} \right)$$

$$\delta_B = 17wL^4/256EI$$



5. THE CONJUGATE BEAM METHOD

Worked Example 5.5

Determine the deflection δ_C and slope θ_A for the beam below, in terms of w , L and EI . Assume $EI = \text{constant}$.

Solution

The slope at A is equal to the shear force at A on the conjugate beam. Thus :

$$\theta_A = \left(\frac{2}{3} \times \frac{L}{2} \times \frac{wL^2}{8EI} \right) = \frac{wL^3}{24EI}$$

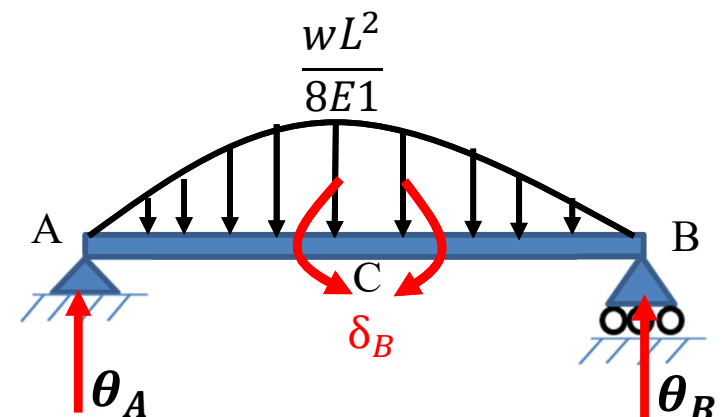
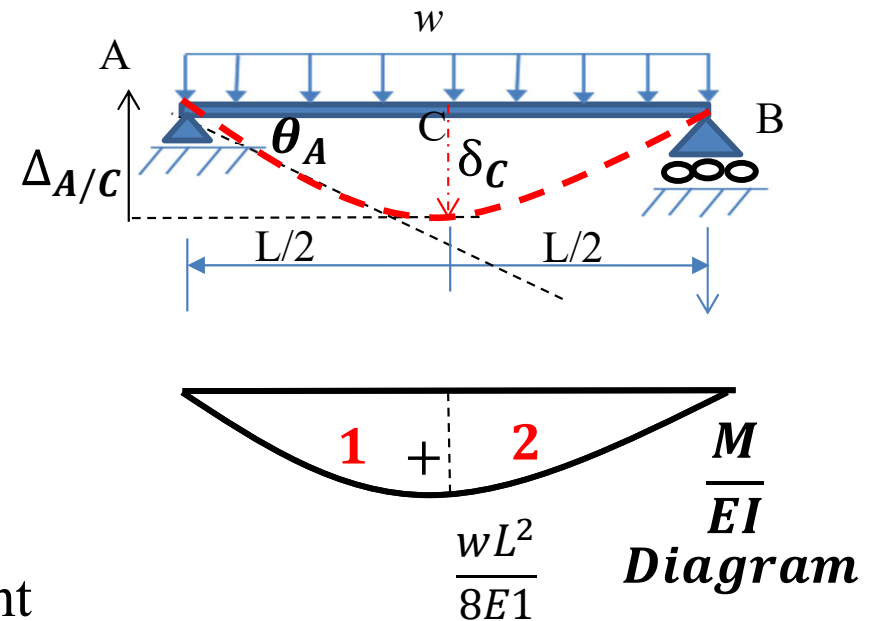
$$\theta_A = \frac{wL^3}{24EI}$$

The deflection at C is equal to the moment at C on the conjugate beam. Thus:

$$\delta_C = \left(\theta_A \times \frac{L}{2} \right) - \left\{ \left(\frac{2}{3} \times \frac{L}{2} \times \frac{wL^2}{8EI} \right) \times \left(\frac{3}{8} \times \frac{L}{2} \right) \right\}$$

$$\delta_C = \left(\frac{wL^3}{24EI} \times \frac{L}{2} \right) - \left(\frac{wL^3}{24EI} \times \frac{3L}{16} \right) = \frac{5wL^4}{384EI}$$

$$\delta_C = 5wL^4/384EI$$



5. THE CONJUGATE BEAM METHOD

Worked Example 5.6

Using the Conjugate beam method, evaluate θ_A and δ_C for the beam below, in terms of P, L and EI. Assume EI = Varies as shown

Solution

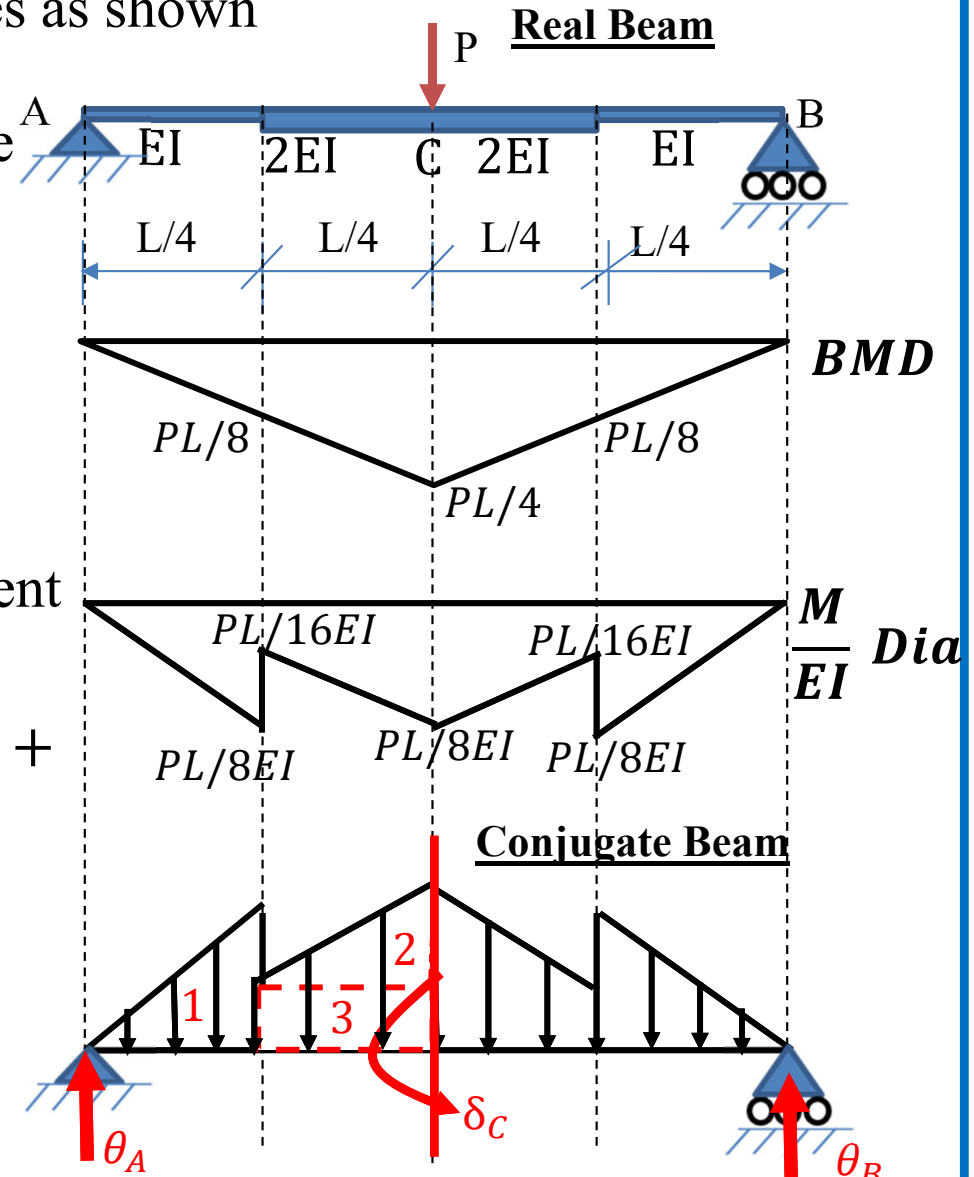
The slope at A is equal to the shear force at A on the conjugate beam. Thus :

$$\theta_A = \left(\frac{1}{2} \times \frac{L}{4} \times \frac{PL}{8EI} \right) + \left(\frac{1}{2} \times \frac{L}{4} \times \frac{PL}{16EI} \right) + \left(\frac{L}{4} \times \frac{PL}{16EI} \right) \therefore \theta_A = \frac{5PL^2}{128EI}$$

The deflection at C is equal to the moment at C on the conjugate beam.

$$\delta_C = \left(\theta_A \times \frac{L}{2} \right) - \left\{ \left(A_1 \times \frac{L}{3} \right) + \left(A_2 \times \frac{L}{12} \right) + \left(A_3 \times \frac{L}{8} \right) \right\}$$

$$\delta_C = \left(\frac{5PL^2}{128EI} \times \frac{L}{2} \right) - \left\{ \left(\frac{PL^2}{64EI} \times \frac{L}{3} \right) + \left(\frac{PL^2}{128EI} \times \frac{L}{12} \right) + \left(\frac{PL^2}{64EI} \times \frac{L}{8} \right) \right\} = \frac{3PL^3}{256EI}$$

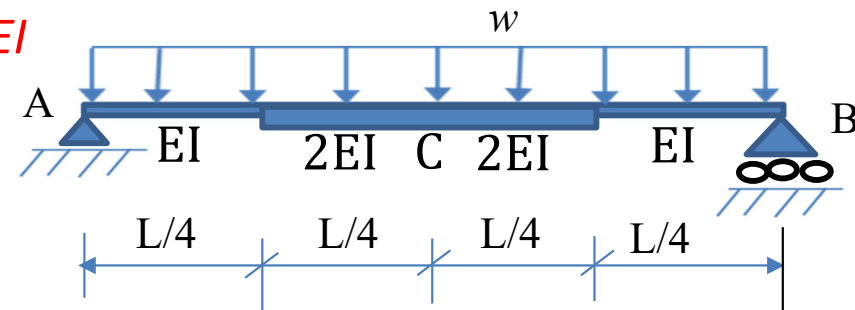


5. THE CONJUGATE BEAM METHOD

Exercise 5.1

Using the conjugate beam method, determine the deflection δ_C and slope θ_A for the beam below, in terms of w , L and EI . Assume EI = Varies as shown

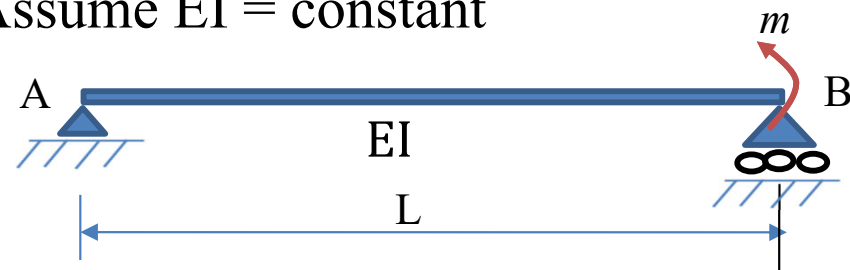
$$\delta_C = 31wL^4 / 4096EI, \theta_A = 7wL^3 / 256EI$$



Exercise 5.2

Using the conjugate beam method, determine the slopes θ_A and θ_B for the beam below, in terms of m , L and EI . Assume EI = constant

$$\theta_A = -mL / 6EI, \theta_B = mL / 3EI$$

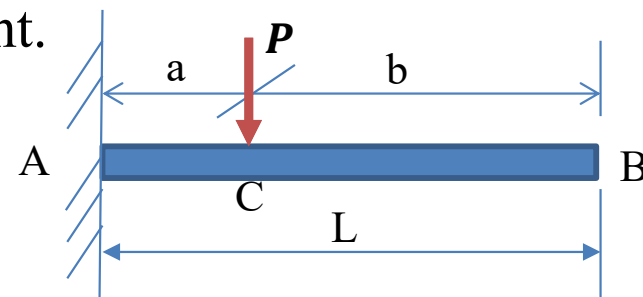


Exercise 5.3

Using the conjugate beam method, evaluate δ_C , δ_B , θ_C & θ_B for beam below, in terms of P , L , a , b & EI . Assume EI = constant.

$$\delta_C = Pa^3 / 3EI, \theta_C = Pa^2 / 2EI$$

$$\delta_B = Pa^2 \{ 3(a+b) - a \} / 6EI, \theta_B = Pa^2 / 2EI$$



6. VIRTUAL FORCE (UNIT LOAD) METHOD

6.1: Introduction

- The methods covered so far (Direct Integration, Macaulay's, Conjugate beam, Moment-Area) are only applicable to normal beams subject to normal loading. If the structure has members which lie in more than one longitudinal axis, such as in frames and trusses, these methods becomes tedious to use.
- The Virtual force method (sometimes referred to as Energy Methods) is applicable not only to beams and rigid jointed frames, but also to trusses, hybrid frames and even arches. The method is therefore one of the most powerful and versatile tools in structural analysis.

Basis

- The Virtual force method is based on the principle of conservation of energy which was developed by Daniel Bernoulli © 1750, based on the principle of virtual work, which was developed by Johann Bernoulli © 1715.
- **Virtual work** may be defined as the work done by a real force system acting on a body, which is in static equilibrium, when the force system is given some **virtual displacement**.

6. VIRTUAL FORCE (UNIT LOAD) METHOD

- “**Virtual**” means imaginary, or fictitious, or unreal. The term **displacement** means a change in position, or a change in orientation. The term **deflection** is used to refer to the change in position, while the term **slope** is used to refer to the change in orientation (or rotation).
- **Virtual work** may also be defined as the work done by a **virtual force** system acting on a body, which is in static equilibrium, when the force system is given some **real displacement**.
- The principle of conservation of energy then dictates that “For a system of forces acting on a body, the body is in static equilibrium if, when it is given any virtual displacement, the net work done by the system of forces is zero”. Thus, **External work done = Internal work done**.

Internal Work

- In flexural structures (i.e. beams and rigid-jointed frames), the internal work done is stored as strain energy due to flexural, shear, and axial strains. However, the strain energy due to shear and axial strains is negligible compared to the strain energy due to flexural strains. Therefore, only flexural strains are considered in determining displacements in flexural structures.

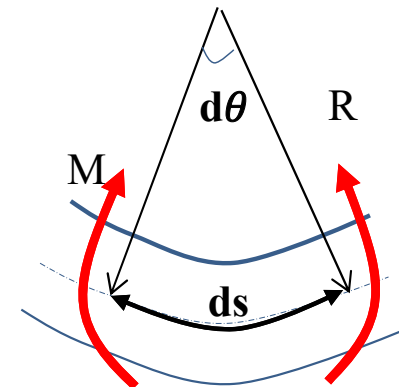
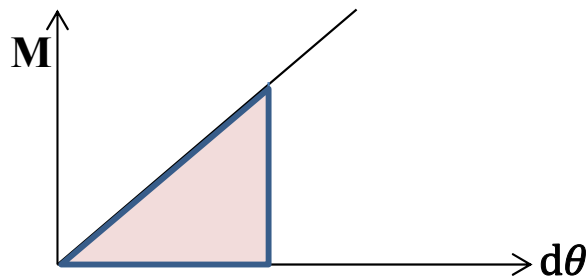
6. VIRTUAL FORCE (UNIT LOAD) METHOD

- In trusses, the members are subjected to axial forces only, and the internal work done is stored as strain energy due to axial strains. Therefore, only axial strains are considered in determining displacements in trusses.
- The displacements in hybrid frames are determined by considering axial strains in the truss-type members and flexural strains in the beam-type members.

6.2 BEAM DISPLACEMENTS

Consider an isolated section of beam subjected to bending moment as shown in the sketch below

The internal work done on the beam element ds by a bending moment M causing a rotation $d\theta$;



As the bending moment increases, the angle of rotation also increases linearly as shown

6. VIRTUAL FORCE (UNIT LOAD) METHOD

$$\text{Internal work done } dw_i = \frac{1}{2} M \cdot d\theta$$

$$ds = R \cdot d\theta \quad \text{Or } d\theta = \frac{ds}{R} \quad \text{Or } \frac{1}{R} = \frac{d\theta}{ds}$$

$$\text{From beam theory, } \frac{M}{EI} = \frac{1}{R}$$

$$\therefore \frac{d\theta}{ds} = \frac{M}{EI} \Rightarrow d\theta = \frac{M}{EI} \cdot ds$$

$$\therefore dw_i = \frac{1}{2} M \cdot d\theta = \frac{1}{2} M \cdot \frac{M}{EI} \cdot ds = \frac{M^2}{2EI} ds$$

$$\text{Internal work done on the whole beam } W_i = \int_0^L \frac{M^2}{2EI} ds$$

But since θ is usually small, $ds \simeq dx$

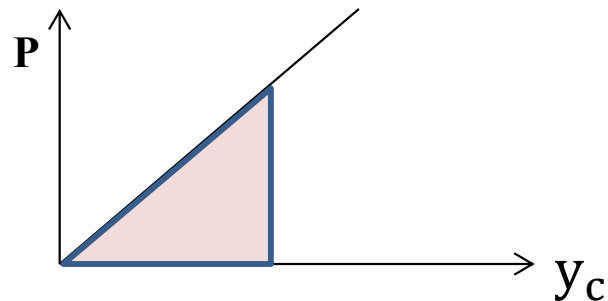
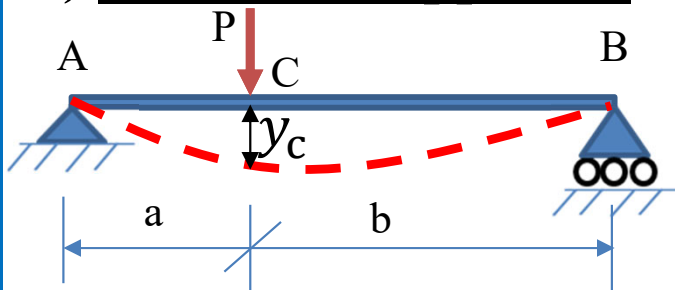
$$\therefore W_i = \int_0^L \frac{M_x^2}{2EI} dx ; \text{ This work is stored as strain energy (U) in the beam}$$

$$W_i = U = \int_0^L \frac{M_x^2}{2EI} dx$$

The strain energy due to bending can be used to determine deflection of the beam in three ways:

6. VIRTUAL FORCE (UNIT LOAD) METHOD

1.) Real Work Approach



External Work

$$\text{External work done } W_e = \frac{1}{2} P \cdot y_c$$

Total external work = total strain energy
(internal work)

$$\Rightarrow W_e = W_i$$

$$\Rightarrow \frac{1}{2} P \cdot y_c = \int_0^L \frac{M_x^2}{2EI} dx$$

$$\text{Or } y_c = \int_0^L \frac{M_x}{PEI} dx$$

The real work approach can only be used to determine deflection under a single point load:

6. VIRTUAL FORCE (UNIT LOAD) METHOD

2) Castigliano's First Theorem

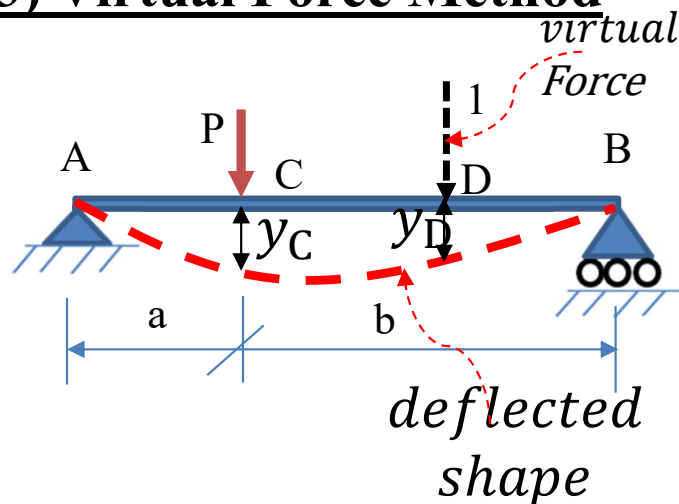
The theorem states that, if the total strain energy of a structure is differentiated with respect to any of the applied forces then the result is the displacement at the point of action of the force along its line of action

$$\Rightarrow U = \frac{1}{2EI} \int_0^L M_x^2 \cdot dx$$

$$\frac{dU}{dP} = \frac{1}{2EI} \int_0^L 2M_x \left(\frac{dM_x}{dP} \right) \cdot dx = y_c$$

The Castigliano's approach is versatile in loading but deflection can only be obtained at points of application of loads

3) Virtual Force Method



External work done by the applied force

$$W_P = \frac{1}{2} P \cdot y_c$$

External work done by the Imaginary unit force at D = $W_I = 1 \cdot y_D$

Total External work done $W_e = W_P + W_I$

$$\Rightarrow W_e = \frac{1}{2} P \cdot y_c + y_D$$

6. VIRTUAL FORCE (UNIT LOAD) METHOD

Internal work done $U_P = \frac{1}{2EI} \int_0^L M_x^2 \cdot dx$

Internal work done by imaginary unit force causing imaginary bending moment m , $U_i = \int_0^L (m) \cdot d\theta$

$$U_i = \int_0^L m \cdot \frac{M}{EI} \cdot ds = \frac{1}{EI} \int_0^L m_x \cdot M_x \cdot dx$$

Total strain energy stored, $U = U_P + U_i$

$$U = \frac{1}{2EI} \int_0^L M_x^2 \cdot dx + \frac{1}{EI} \int_0^L m_x \cdot M_x \cdot dx$$

Equating external work done to total strain energy stored we get

$$\frac{1}{2} P \cdot y_C + y_D = \frac{1}{2EI} \int_0^L M_x^2 \cdot dx + \frac{1}{EI} \int_0^L m_x \cdot M_x \cdot dx$$

$$\text{But } \frac{1}{2} P \cdot y_C = \frac{1}{2EI} \int_0^L M_x^2 \cdot dx$$

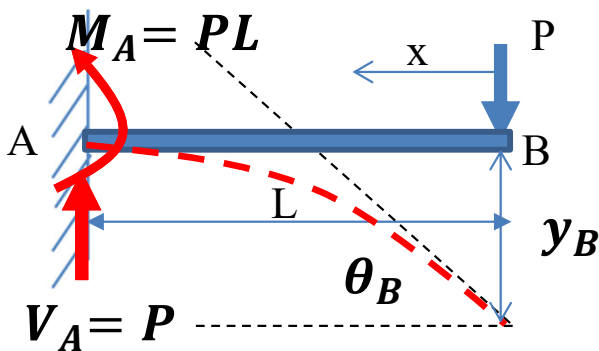
$$\therefore y_D = \frac{1}{EI} \int_0^L m_x \cdot M_x \cdot dx$$

The Virtual load approach is the most versatile in loading and deflection/slope can be obtained at any given point along the beam

6. VIRTUAL FORCE (UNIT LOAD) METHOD

Application of Real Work Approach

Worked Example 6.1



$$M_x = Px \text{ and } M_x^2 = (Px)^2$$

$$\text{Internal work done } U = \frac{1}{2EI} \int_0^L M_x^2 \cdot dx$$

$$\text{Or } U = \frac{1}{2EI} \int_0^L (Px)^2 \cdot dx$$

$$U = \frac{1}{2EI} \int_0^L P^2 x^2 \cdot dx = \frac{1}{2EI} \left[\frac{P^2 x^3}{3} \right]_0^L = \frac{P^2 L^3}{6EI}$$

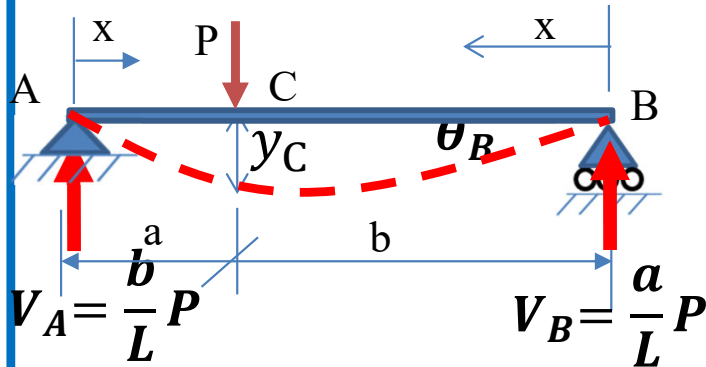
$$\text{External work done by the applied force} = W_P = \frac{1}{2} P \cdot y_B$$

$$\text{Equating external work to internal work} \approx \frac{1}{2} P \cdot y_B = \frac{P^2 L^3}{6EI}$$

$$\Rightarrow y_B = \frac{PL^3}{3EI}$$

6. VIRTUAL FORCE (UNIT LOAD) METHOD

Worked Example 6.2



Section AC

Origin at A; $M_x = \frac{b}{L} Px$

Internal work done $U_{AC} = \frac{1}{2EI} \int_0^a \left(\frac{b}{L} Px\right)^2 \cdot dx$

$$U_{AC} = \frac{1}{2EI} \int_0^a \frac{b^2}{L^2} P^2 x^2 \cdot dx$$

$$U_{AC} = \frac{1}{2EI} \left[\frac{b^2}{3L^2} P^2 x^3 \right]_0^a = \frac{P^2 b^2 a^3}{6EIL^2}$$

Section BC

Origin at B; $M_x = \frac{a}{L} Px$

Internal work done $U_{BC} = \frac{1}{2EI} \int_0^b \left(\frac{a}{L} Px\right)^2 \cdot dx$, $U_{BC} = \frac{1}{2EI} \int_0^b \frac{a^2}{L^2} P^2 x^2 \cdot dx$

$$U_{BC} = \frac{1}{2EI} \left[\frac{a^2}{3L^2} P^2 x^3 \right]_0^b = \frac{P^2 a^2 b^3}{6EIL^2}$$

Total strain energy stored $U = U_{AC} + U_{BC} = \frac{P^2 a^3 b^2}{6EIL^2} + \frac{P^2 a^2 b^3}{6EIL^2} = \frac{P^2 a^2 b^2 (a+b)}{6EIL^2}$

External work done by the applied force = $W_P = \frac{1}{2} P \cdot y_C$

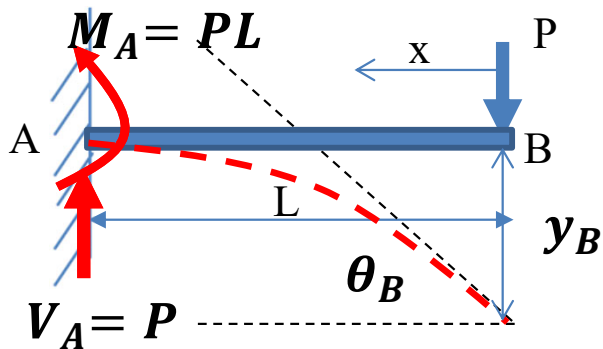
Equating external work to internal work $\simeq \frac{1}{2} P \cdot y_C = \frac{P^2 a^2 b^2 (a+b)}{6EIL^2}$

But $(a+b) = L \therefore \Rightarrow y_C = \frac{Pa^2 b^2}{3EIL}$

6. VIRTUAL FORCE (UNIT LOAD) METHOD

Application of Castigliano's First Theorem

Worked Example 6.3



$$M_x = Px$$

$$y_B = \frac{dU}{dP} = \frac{1}{EI} \int_0^L M_x \left(\frac{dM_x}{dP} \right) \cdot dx$$

$$\frac{dM_x}{dP} = x$$

$$\therefore y_B = \frac{1}{EI} \int_0^L Px(x) \cdot dx$$

$$y_B = \frac{P}{EI} \int_0^L x^2 \cdot dx = \frac{P}{EI} \left[\frac{x^3}{3} \right]_0^L = \frac{PL^3}{3EI}$$

6. VIRTUAL FORCE (UNIT LOAD) METHOD

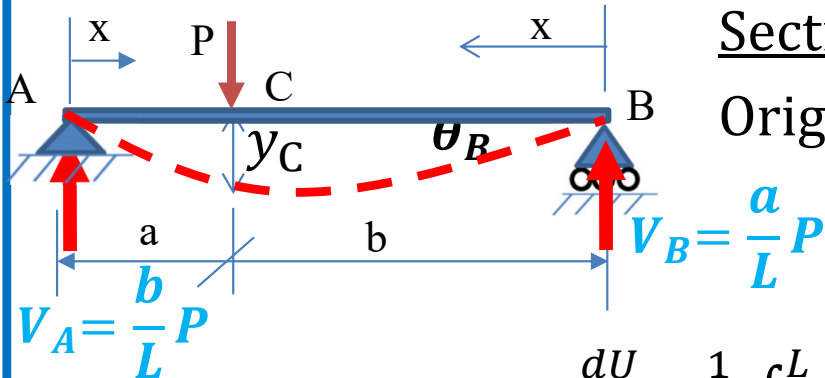
Worked Example 6.4

Section AC

Origin at A; $M_x = \frac{b}{L}Px$, and $\frac{dM_x}{dP} = \frac{b}{L}x$

Section BC

Origin at B; $M_x = \frac{a}{L}Px$, and $\frac{dM_x}{dP} = \frac{a}{L}x$



$$y_C = \frac{dU}{dP} = \frac{1}{EI} \int_0^L M_x \left(\frac{dM_x}{dP} \right) \cdot dx$$

$$y_C = \frac{1}{EI} \int_0^a M_x \left(\frac{dM_x}{dP} \right) \cdot dx + \frac{1}{EI} \int_0^b M_x \left(\frac{dM_x}{dP} \right) \cdot dx$$

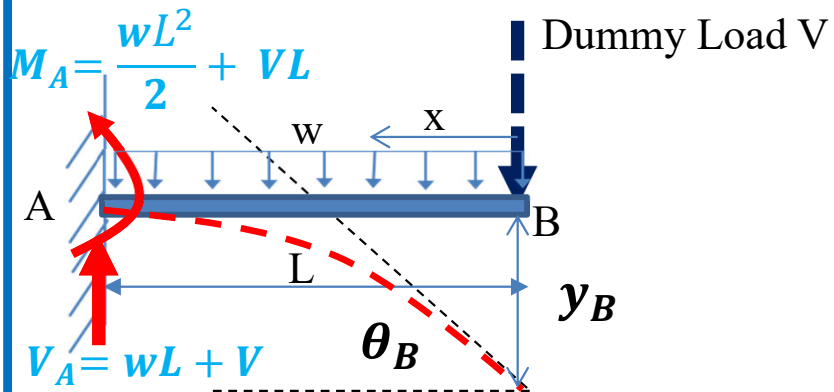
$$y_C = \frac{1}{EI} \int_0^a \left(\frac{b}{L}Px \right) \left(\frac{b}{L}x \right) \cdot dx + \frac{1}{EI} \int_0^b \left(\frac{a}{L}Px \right) \left(\frac{a}{L}x \right) \cdot dx$$

$$y_C = \frac{Pb^2}{EIL^2} \int_0^a x^2 \cdot dx + \frac{Pa^2}{EIL^2} \int_0^b x^2 \cdot dx$$

$$y_C = \frac{Pb^2a^3}{3EIL^2} + \frac{Pa^2b^3}{3EIL^2} = \frac{Pb^2a^2(a+b)}{3EIL^2} \quad \text{But } (a+b) = L \therefore y_C = \frac{Pb^2a^2}{3EIL}$$

6. VIRTUAL FORCE (UNIT LOAD) METHOD

Worked Example 6.5(cantilever with udl)



With origin at B, $M_x = \frac{wx^2}{2} + Vx$ and $\frac{dM_x}{dV} = x$

$$y_B = \frac{dU}{dV} = \frac{1}{EI} \int_0^L M_x \left(\frac{dM_x}{dV} \right) \cdot dx = \frac{1}{EI} \int_0^L \left(\frac{wx^2}{2} + Vx \right) (x) \cdot dx$$

$$y_B = \frac{1}{EI} \int_0^L \left(\frac{wx^3}{2} + Vx^3 \right) dx$$

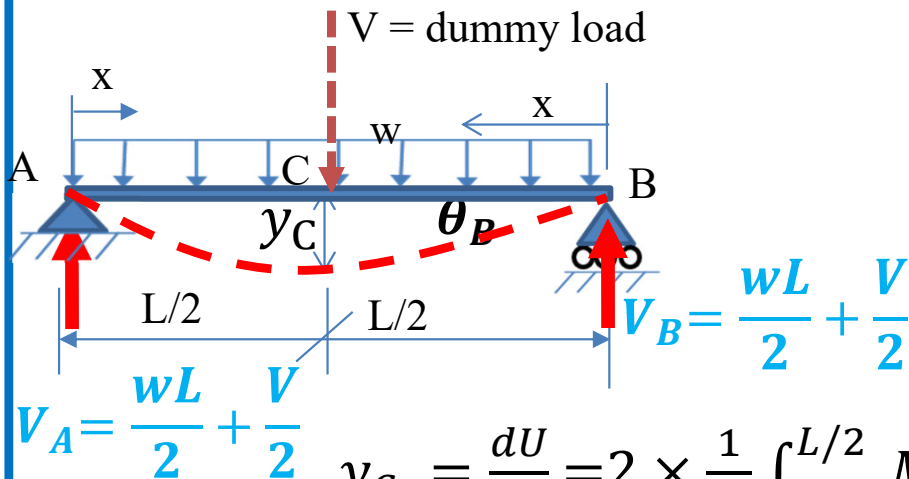
$$y_B = \frac{1}{EI} \left[\frac{wx^4}{8} + \frac{Vx^3}{3} \right]_0^L = \frac{1}{EI} \left(\frac{wL^4}{8} + \frac{VL^3}{3} \right)$$

But $V=0$

$$y_B = \frac{wL^4}{8EI}$$

6. VIRTUAL FORCE (UNIT LOAD) METHOD

Worked Example 6.6



Section AC

Origin at A; $M_x = \left(\frac{wL}{2} + \frac{V}{2}\right) \cdot x - \frac{wx^2}{2}$,

and $\frac{dM_x}{dV} = \frac{x}{2}$

Section BC

Origin at B; SAME AS FOR SECTION AC

$$y_C = \frac{dU}{dV} = 2 \times \frac{1}{EI} \int_0^{L/2} M_x \left(\frac{dM_x}{dV} \right) \cdot dx$$

$$y_C = 2 \times \frac{1}{EI} \int_0^{L/2} M_x \left(\frac{dM_x}{dV} \right) \cdot dx$$

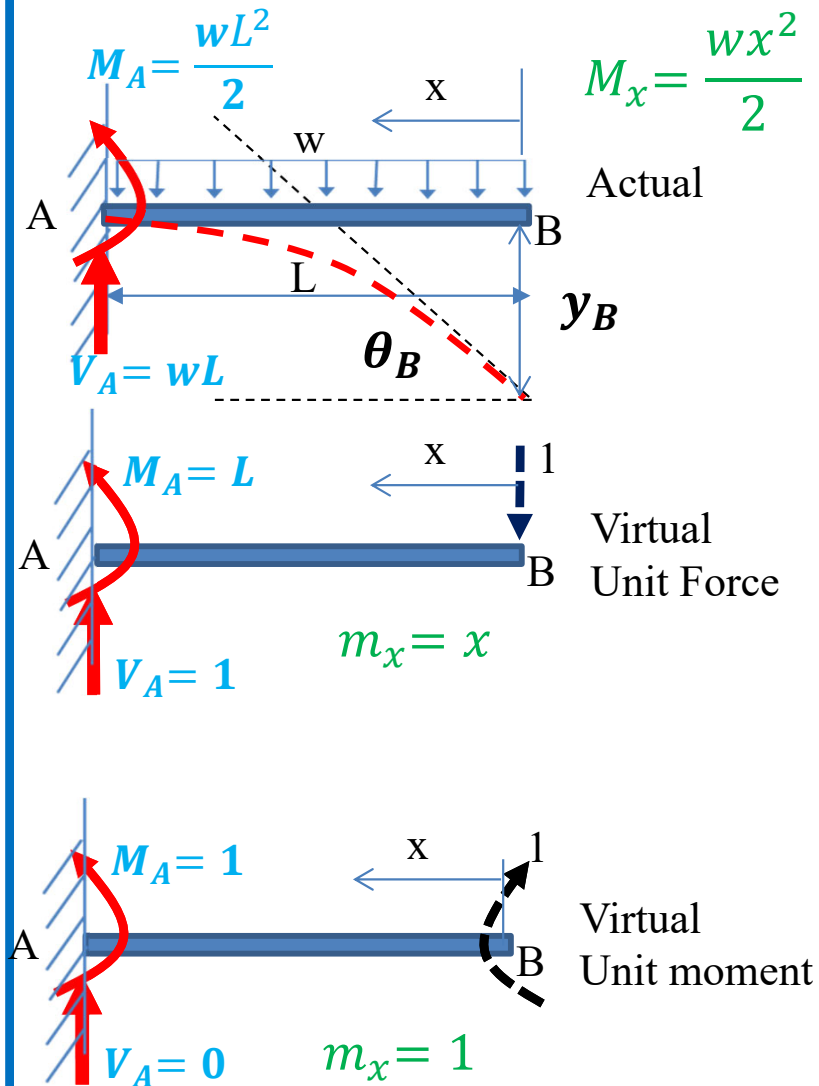
$$y_C = \frac{2}{EI} \int_0^{L/2} \left(\frac{wLx}{2} + \frac{Vx}{2} - \frac{wx^2}{2} \right) (x/2) dx = \frac{2}{EI} \int_0^{L/2} \left(\frac{wLx^2}{4} + \frac{Vx^2}{4} - \frac{wx^3}{4} \right) \cdot dx$$

$$y_C = \frac{2}{EI} \left(\frac{wLx^3}{12} + \frac{Vx^3}{12} - \frac{wx^4}{16} \right) \Big|_0^{L/2} = \frac{2}{EI} \left(\frac{wL^4}{96} + \frac{VL^3}{96} - \frac{wL^4}{256} \right)$$

$$\text{But } V = 0 \therefore y_C = \frac{2}{EI} \left(\frac{wL^4}{96} - \frac{wL^4}{256} \right) = \frac{2}{EI} \left(\frac{(8-3)wL^4}{768} \right) = \frac{5wL^4}{384EI}$$

6. VIRTUAL FORCE (UNIT LOAD) METHOD

Application of Unit Load Method



Worked Example 6.7

$$1. y_B = \frac{1}{EI} \int_0^L m_x \cdot M_x \cdot dx$$

$$y_B = \frac{1}{EI} \int_0^L x \cdot \left(\frac{wx^2}{2}\right) \cdot dx = \frac{1}{EI} \int_0^L \frac{wx^3}{2} \cdot dx$$

$$y_B = \frac{1}{EI} \left[\frac{wx^4}{8} \right]_0^L = \frac{1}{EI} \left(\frac{wL^4}{8} \right)$$

$$y_B = \frac{wL^4}{8EI}$$

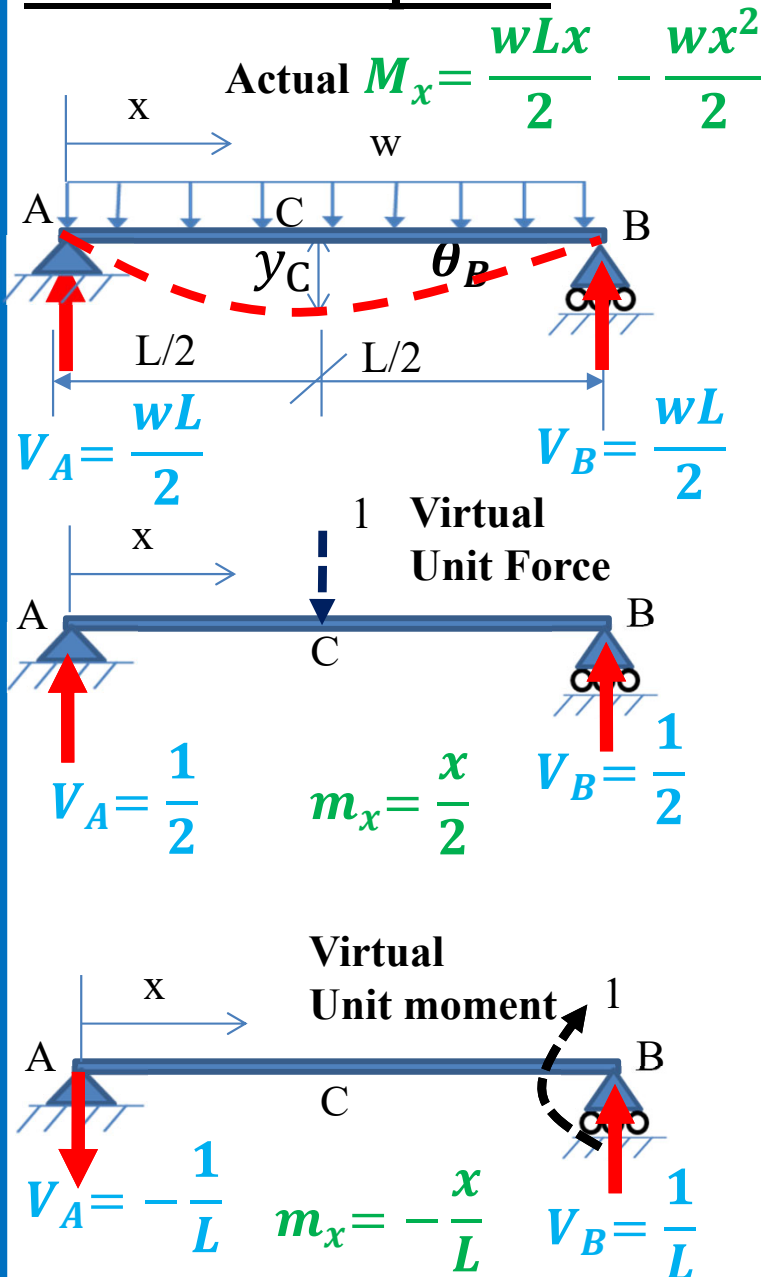
$$1. \theta_B = \frac{1}{EI} \int_0^L m_x \cdot M_x \cdot dx$$

$$\theta_B = \frac{1}{EI} \int_0^L 1 \cdot \left(\frac{wx^2}{2}\right) \cdot dx = \frac{1}{EI} \left[\frac{wx^3}{6} \right]_0^L$$

$$\theta_B = \frac{wL^3}{6EI}$$

6. VIRTUAL FORCE (UNIT LOAD) METHOD

Worked Example 6.8



$$1. y_C = \frac{1}{EI} \int_0^L m_x \cdot M_x \cdot dx$$

$$y_C = 2 \times \frac{1}{EI} \int_0^{L/2} \frac{x}{2} \cdot \left(\frac{wLx}{2} - \frac{wx^2}{2} \right) \cdot dx$$

$$y_C = \frac{2}{EI} \int_0^{L/2} \left[\frac{wLx^2}{4} - \frac{wx^3}{4} \right] dx$$

$$y_C = \frac{2}{EI} \left[\frac{wLx^3}{12} - \frac{wx^4}{16} \right]_0^{L/2} = \frac{2}{EI} \left[\frac{wL^4}{96} - \frac{wL^4}{256} \right]$$

$$y_C = \frac{2}{EI} \left[\frac{(8-3)wL^4}{768} \right] = \frac{5wL^4}{384EI}$$

$$1. \theta_B = \frac{1}{EI} \int_0^L m_x \cdot M_x \cdot dx$$

$$\theta_B = \frac{1}{EI} \int_0^L -\frac{x}{L} \cdot \left(\frac{wLx}{2} - \frac{wx^2}{2} \right) \cdot dx$$

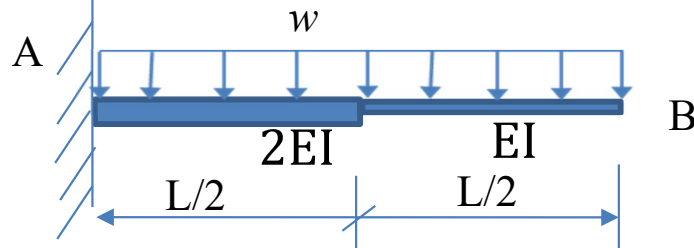
$$\theta_B = \frac{1}{EI} \int_0^L \left(-\frac{wx^2}{2} + \frac{wx^3}{2L} \right) \cdot dx = \frac{1}{EI} \left[-\frac{wx^3}{6} + \frac{wx^4}{8L} \right]_0^L$$

$$\theta_B = -\frac{wL^3}{24EI}$$

6. VIRTUAL FORCE (UNIT LOAD) METHOD

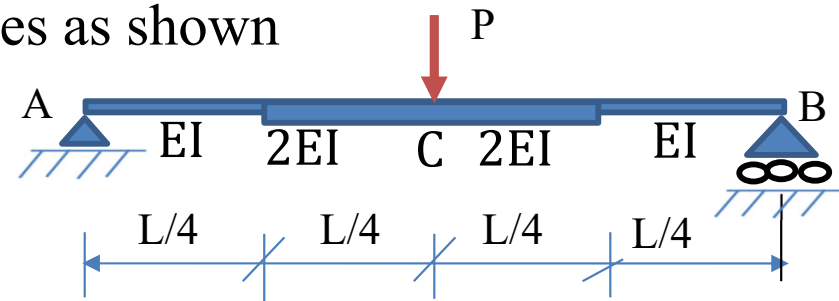
Exercise 6.1

Using the virtual force method, evaluate the deflection δ_B and slope θ_B for the beam below, in terms of w , L and EI . Assume EI = Varies as shown



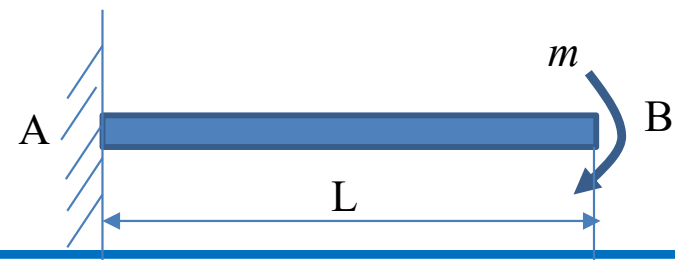
Exercise 6.2

Using the virtual force method, evaluate θ_A and δ_C for the beam below, in terms of P , L and EI . Assume EI = Varies as shown



Exercise 6.3

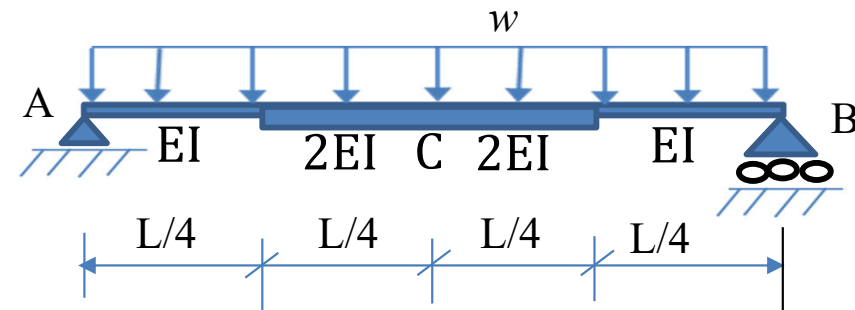
Using the virtual force method, evaluate δ_B & θ_B for beam below, in terms of P , L , a , b & EI . Assume EI = constant.



6. VIRTUAL FORCE (UNIT LOAD) METHOD

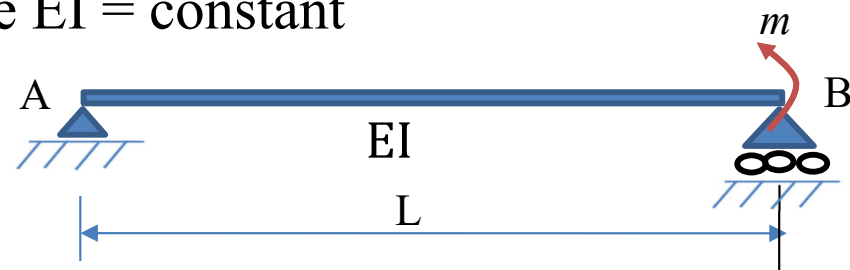
Exercise 6.4

Using the virtual force method, determine the deflection δ_C and slope θ_A for the beam below, in terms of w , L and EI . Assume EI = Varies as shown



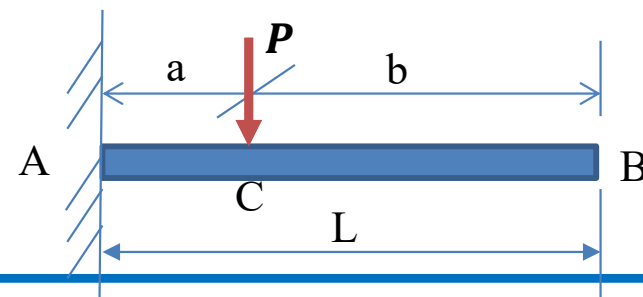
Exercise 6.5

Using virtual force method, determine the slopes θ_A and θ_B for the beam below, in terms of m , L and EI . Assume EI = constant



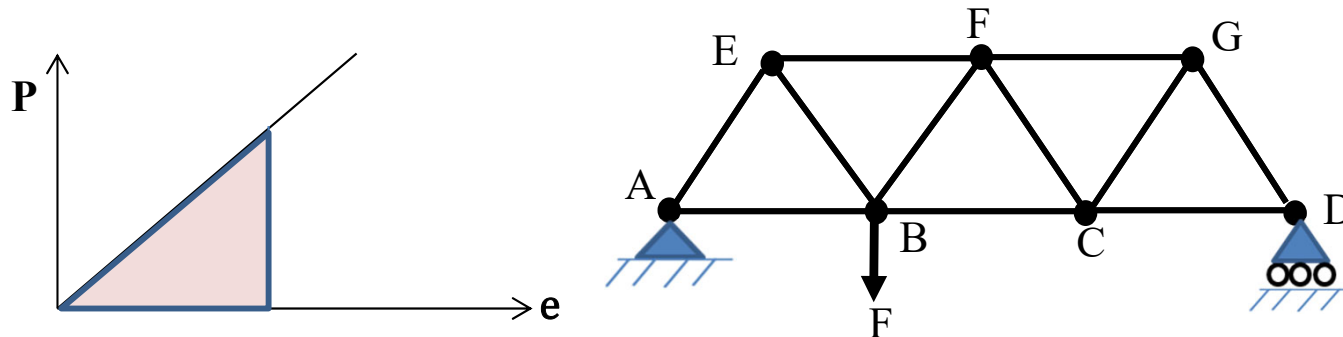
Exercise 6.6

Using virtual force method, evaluate δ_C , δ_B , θ_C & θ_B for beam below, in terms of P , L , a , b & EI . Assume EI = constant.



7 TRUSS DISPLACEMENT

- Trusses are pin-jointed frames and usually loaded at the joints. When a truss is loaded, each member will be subjected to a tensile or compressive force.
- As a result of these forces, each member will suffer elastic strain leading to a change in the original length of the member. The members in tension will become longer while the members in compression become shorter. However, compatibility at the joints requires that the members must remain joined together.
- In order to satisfy equilibrium and compatibility the joints which are not restrained will be displaced from their original positions, so that the truss assumes a slightly different geometry. The geometry adopted by the truss after loading is referred to as the deformed shape. The member elongations or shortenings are usually very small in comparison to their original lengths. However, the cumulative effect of the member straining can lead to a significant change in the original geometry of the truss.



- Structural analysis aims at predicting and evaluating the joint displacements since they affect certain design criteria.

7 TRUSS DISPLACEMENT

Internal work done on one truss member $w_i = \frac{1}{2} P \cdot e$

But young's modulus(E) = Stress(σ)/strain (ϵ)

$$E = \sigma / \epsilon, \epsilon = e / L, \text{ and } \sigma = P / A \quad \therefore E = P / A \times L / e$$

$$\Rightarrow e = PL / AE \quad \therefore w_i = \frac{1}{2} P \cdot e = \frac{1}{2} P \times PL / AE = P^2 L / 2AE$$

Total strain Energy stored, $U = \sum P^2 L / 2AE$

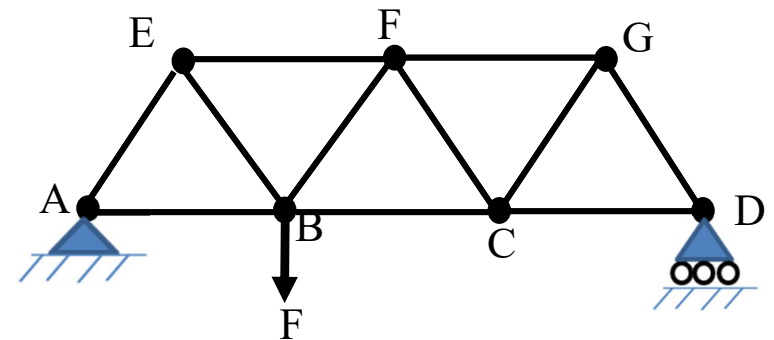
Real Work Approach

External work done $W_e = \frac{1}{2} F \cdot \delta_B$

Internal work done $W_i = \sum P^2 L / 2AE$

Equating Internal work to external work

$$\delta_B = \frac{1}{F} \sum P^2 L / AE$$



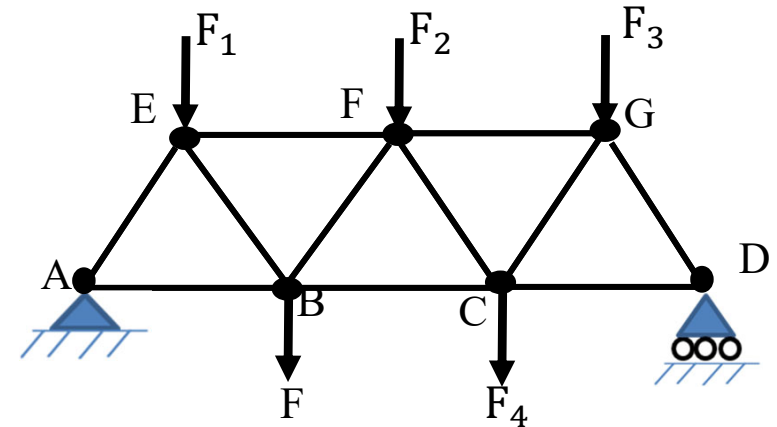
7 TRUSS DISPLACEMENT

Castiglianos 1st Theorem

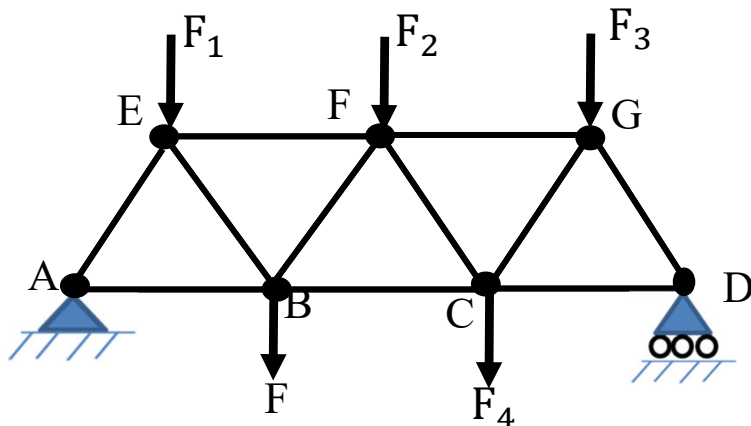
$$\delta_B = \frac{dU}{dF}$$

$$\delta_B = \frac{d \sum P^2 L / 2AE}{dF} = \sum \frac{2PL}{2AE} \cdot \frac{dP}{dF}$$

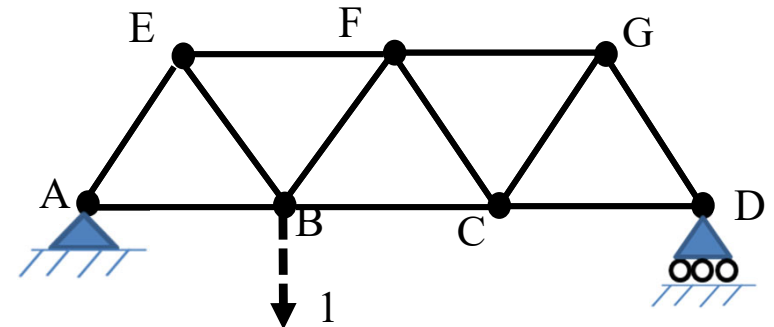
$$\delta_B = \sum \frac{PL}{AE} \cdot \frac{dP}{dF}$$



Virtual Force Method



Actual ; giving actual axial forces, P



Virtual ; giving virtual axial forces, p

$$E.W.D, W_e = \frac{1}{2} F \cdot \delta_B + \frac{1}{2} F_1 \delta_E + \frac{1}{2} F_2 \delta_F + \frac{1}{2} F_3 \delta_G + \frac{1}{2} F_4 \delta_C + 1 \cdot \delta_B$$

7 TRUSS DISPLACEMENT

$$E.W.D, W_e = \frac{1}{2} F \cdot \delta_B + \frac{1}{2} F_1 \delta_E + \frac{1}{2} F_2 \delta_F + \frac{1}{2} F_3 \delta_G + \frac{1}{2} F_4 \delta_C + 1 \cdot \delta_B$$

$$E.W.D, W_e = \sum \frac{1}{2} F \cdot \delta + \delta_B$$

$$I.W.D, W_i = \sum \frac{P^2 L}{2AE} + \sum p \cdot \frac{PL}{AE}$$

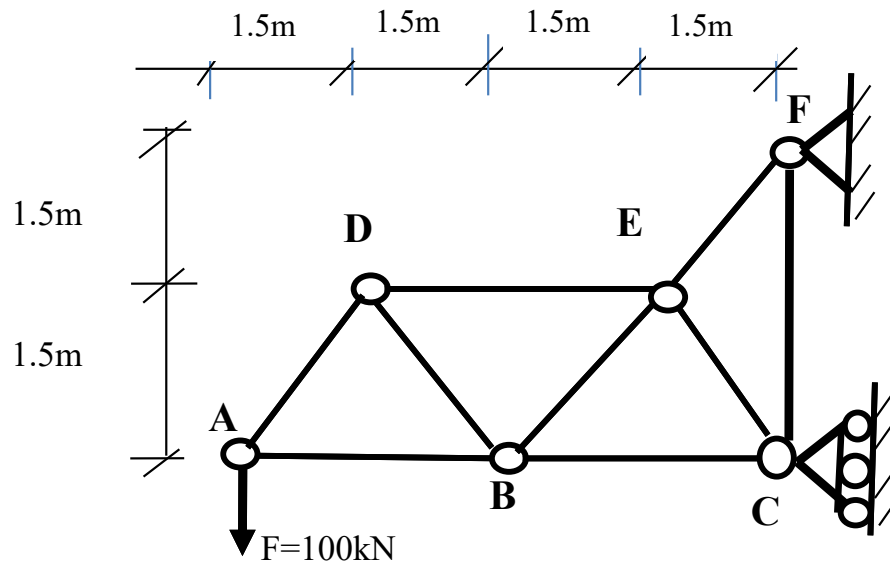
Equating external to internal work

$$\cancel{\sum \frac{1}{2} F \cdot \delta} + \delta_B = \cancel{\sum \frac{P^2 L}{2AE}} + \sum p \cdot \frac{PL}{AE}$$

$$\Rightarrow \delta_B = \sum p \cdot \frac{PL}{AE}$$

7 TRUSS DISPLACEMENT

Worked Example 7.1



Given $E = 210 \text{ kN/mm}^2$

Areas of compression members $= 3310 \text{ mm}^2$

Areas of Tension members $= 2120 \text{ mm}^2$

Determine the vertical deflection at A?

Real Work Method

$$\delta_A = \frac{1}{F} \sum P^2 L / AE$$

Solution Procedure:

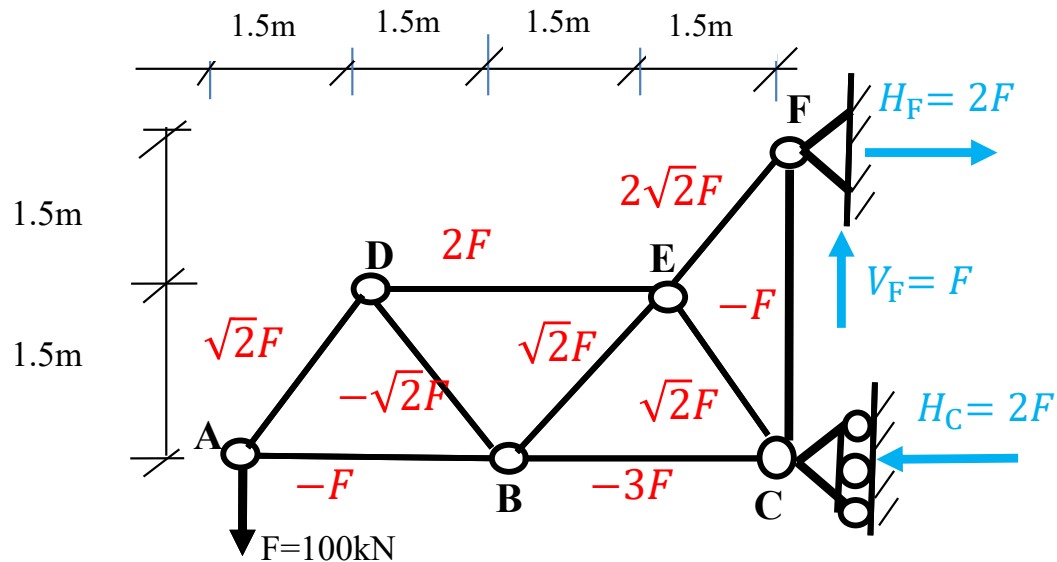
Step 1: Establish whether the member is statically determinate externally and internally; Externally whether the unknown reactions can be solved using the 3 equations of static equilibrium And internally whether $M = 2j - 3$

Step 2: Analyse the truss to establish member internal forces (method of joints or method of section); FCE 201

Step 3: Use equation 1 above to evaluate the deflection

7 TRUSS DISPLACEMENT

Application of Real Work Method



- 3 unknown reactions, 3 equations of static equilibrium hence statically determinate externally
- Number of members $M=9$, number of joints $=6$, and $2j-3=12-3=9$ hence statically determinate internally
- From truss analysis, the member forces are as indicated in the table below

- Sign convention;
Compression = -ve
Tension = +ve

$$\delta_A = \frac{1}{FE} \sum P^2 L / A$$

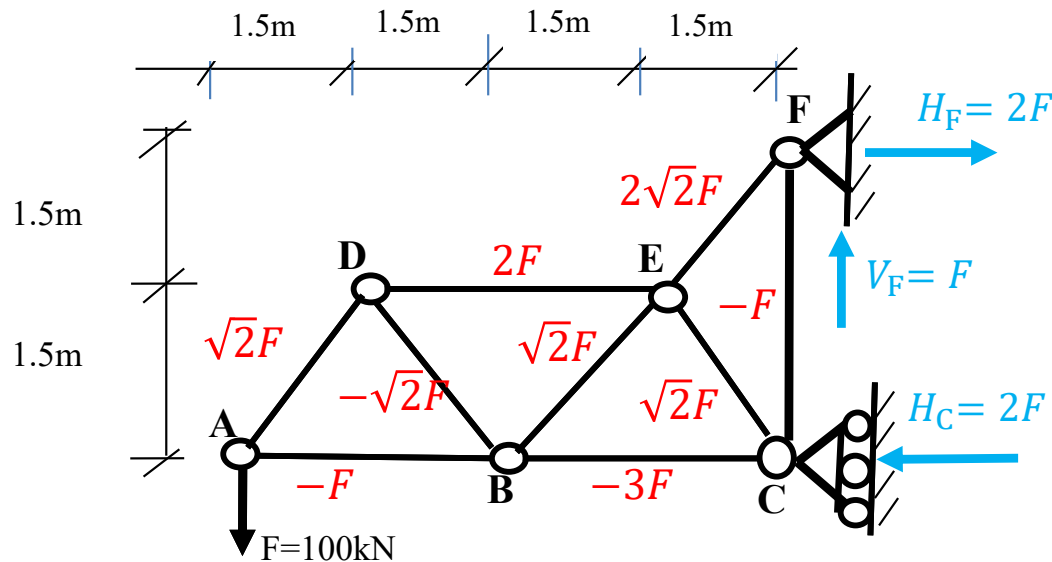
$$\delta_A = \frac{1}{100 \times 210} \times 309180.5$$

$$\delta_A = 14.72 \text{ mm}$$

Member	Length (mm)	Axial Force (kN)	Area(mm ²)	$P^2 L / A$ (kN ² /mm)
AB	3000	-100	3310	9063.4
BC	3000	-300	3310	81571.0
DE	3000	200	2120	56603.8
AD	2121	141.42	2120	20009.1
BD	2121	-141.42	3310	12815.5
BE	2121	141.42	2120	20009.1
CE	2121	141.42	2120	20009.1
EF	2121	282.84	2120	80036.2
CF	3000	-100	3310	9063.4
Σ				309180.5

7 TRUSS DISPLACEMENT

Application of Castigliano's Theorem



$$\delta_A = \sum \frac{PL}{AE} \cdot \frac{dP}{dF}$$

$$\delta_A = \frac{1}{E} \sum \frac{PL}{A} \cdot \frac{dP}{dF}$$

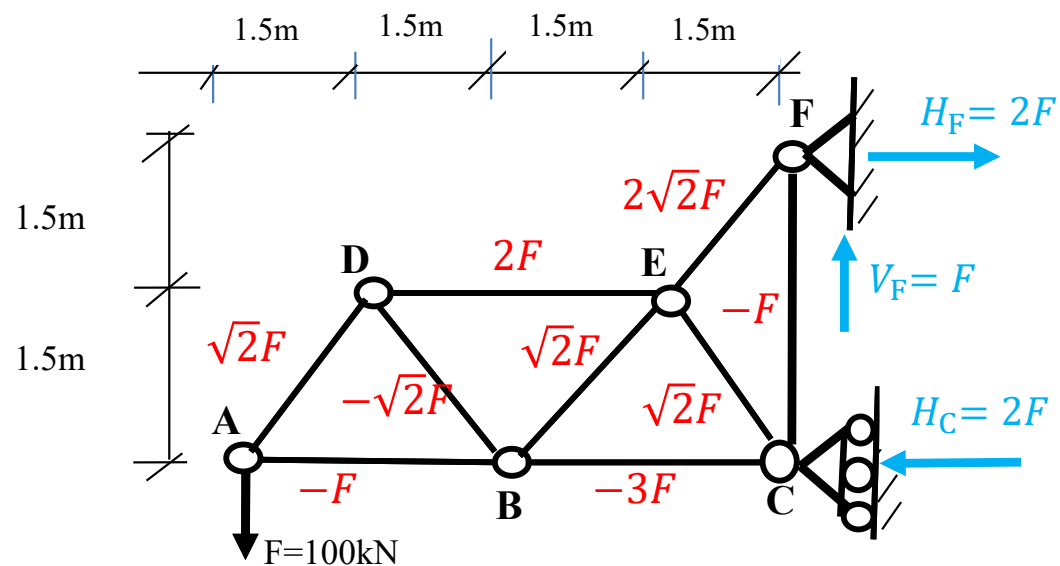
$$\delta_A = \frac{1}{210} \times 3091.8$$

$$\delta_A = 14.72 \text{ mm}$$

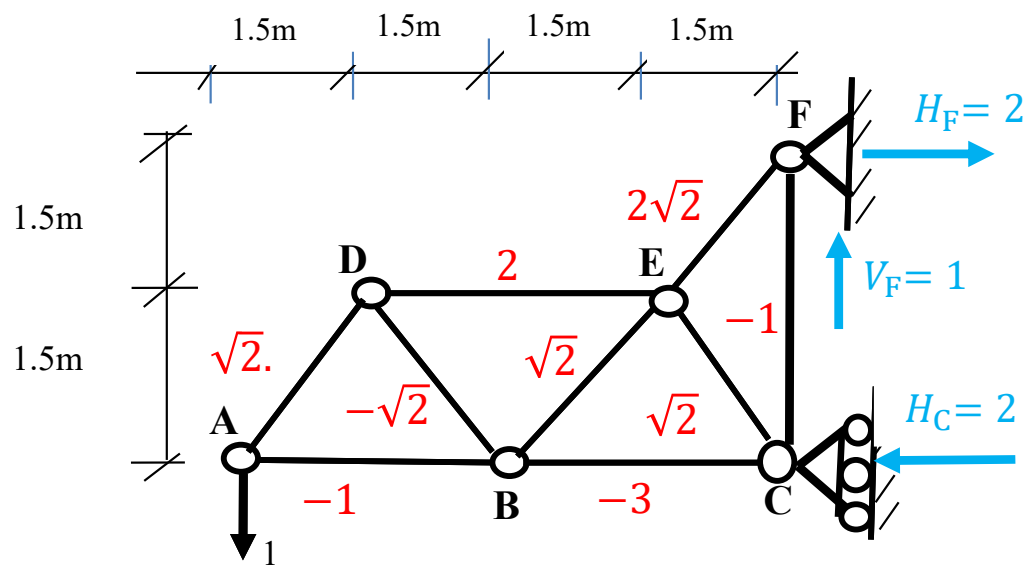
Member	Length, L (mm)	Axial Force, P (kN)	Area, A (mm ²)	dP/dF	$\frac{PL}{A} \cdot \frac{dP}{dF}$ (kN/mm)
AB	3000	-100	3310	-1	90.6
BC	3000	-300	3310	-3	815.7
DE	3000	200	2120	2	566.0
AD	2121	141.42	2120	1.4142	200.1
BD	2121	-141.42	3310	-1.4142	128.2
BE	2121	141.42	2120	1.4142	200.1
CE	2121	141.42	2120	1.4142	200.1
EF	2121	282.84	2120	2.8284	800.4
CF	3000	-100	3310	-1	90.6
				Σ	3091.8

7 TRUSS DISPLACEMENT

Application of Virtual Force Method



ACTUAL



VIRTUAL

7 TRUSS DISPLACEMENT

$$\delta_A = \sum p \cdot \frac{PL}{AE}, \quad \delta_A = \frac{1}{E} \sum p \cdot \frac{PL}{A}$$

Member	Length, L (mm)	Actual Axial Force, P (kN)	Area, A (mm ²)	Virtual Axial Force, <i>p</i> (units)	$p \cdot \frac{PL}{A}$ (kN/mm)
AB	3000	-100	3310	-1	90.6
BC	3000	-300	3310	-3	815.7
DE	3000	200	2120	2	566.0
AD	2121	141.42	2120	1.4142	200.1
BD	2121	-141.42	3310	-1.4142	128.2
BE	2121	141.42	2120	1.4142	200.1
CE	2121	141.42	2120	1.4142	200.1
EF	2121	282.84	2120	2.8284	800.4
CF	3000	-100	3310	-1	90.6
					3091.8

$$\delta_A = \frac{1}{210} \times 3091.8$$

$$\delta_A = 14.72 \text{ mm}$$

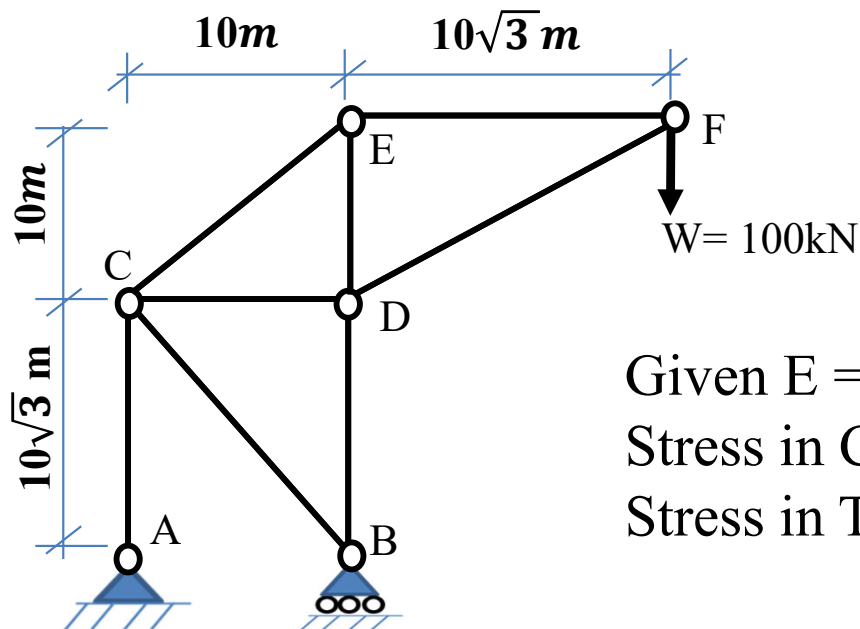
7 TRUSS DISPLACEMENT

Exercise 7.1

For the truss in worked example 7.1, use virtual force method to evaluate the vertical deflection at joint B ($\delta_{B \text{ vertical}}$) and horizontal deflection at joint A ($\delta_{A \text{ horizontal}}$). Compare answers with those of the Castigliano's theorem.

Exercise 7.2

For the truss below, use Castigliano's theorem to evaluate the vertical and horizontal deflection at joint F. What is the magnitude and direction of the resultant deflection at Joint F? Compare results with those of the Virtual force method.



Given $E = 200\text{kN/mm}^2$

Stress in Compression members $\sigma_c = 60\text{N/mm}^2$

Stress in Tension members $\sigma_t = 120\text{N/mm}^2$

7 TRUSS DISPLACEMENT

Exercise 7.3

For the pin jointed frame in Fig. Q 7.3 below has members all of the same material and cross sectional area A except for member BD , which has area kA . Find the value of k , so that when a vertical load is applied at D , the resultant deflection at D is in a direction inclined at 30° to the vertical.

Exercise 7.4

The cross-sectional areas of the truss shown Fig. Q 7.4 are such that due to loading stresses in tension and compression members are 150mpa and 100mpa respectively. Determine the vertical and horizontal deflections of joint C . Take $E = 200\text{kN/mm}^2$

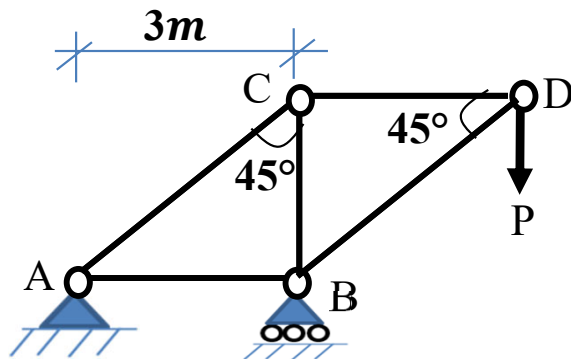


Fig.Q 7.3

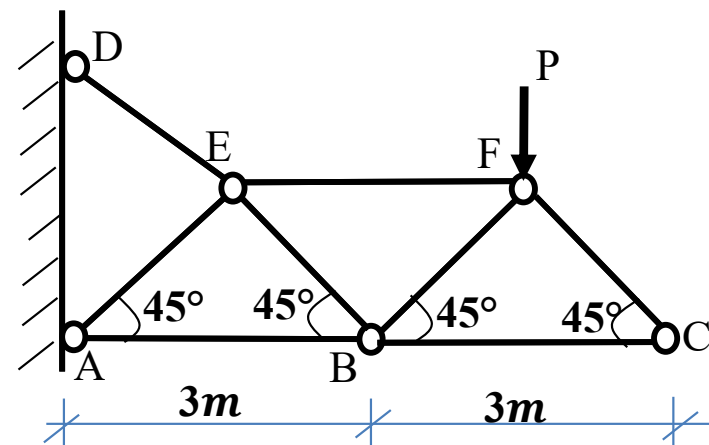


Fig.Q 7.4

8 INFLUENCE LINES

8.1 INTRODUCTION

So far we have studied the effects of applied loads acting at specified points on the structure. In certain structures the applied loads traverse the structure by moving from one end to the other. These moving loads are common in: -

- Bridges: highways, railways, footbridges, etc
- Industrial buildings: overhead cranes, conveyor belts, etc

The moving loads will have both dynamic effects and static effects on the structure. The dynamic effects will be dealt with in FCE 401.

The static effects, namely support reaction components, axial force N , shear force V , and bending moment M , will vary with the position of the load. There is therefore a position at which a given moving load will produce the maximum effects on the structure.

The technique used to study the static effects on the structure as the load moves from one end of the structure to the other is based on the concept of an Influence Line, which was developed by Muller-Breslau (1890).

An influence line is a diagram which graphically represents the effects at a given point on a structure as a unit force occupies different positions on the structure.

The Influence Line for a specific load effect at a specific point on a structure may be plotted using either: -

1. First principles (Equilibrium), OR
2. Second principles (Muller-Breslau Principle)

8 INFLUENCE LINES

8.2 PLOTTING INFLUENCE LINES

8.2.1 Using First Principles

Worked Example 8.1

Plot the I.L.D for reactions V_A and V_B , Shear at C (q_c) and Bending moment at C (M_C) for the beam shown below.

I.L for V_A ; $\Sigma M_B = 0 \Rightarrow V_A L - 1(L-x) = 0$

$$V_A = 1 - x/L$$

I.L for V_B ; $\Sigma M_A = 0 \Rightarrow V_B L - 1(x) = 0$

$$V_B = x/L$$

I.L for q_c :

Load on section AC:

$$q_c = V_A - 1 \Rightarrow q_c = \left(1 - \frac{x}{L}\right) - 1 \Rightarrow q_c = -\frac{x}{L}$$

Load on section CB:

$$q_c = V_A \Rightarrow q_c = \frac{L-x}{L}$$

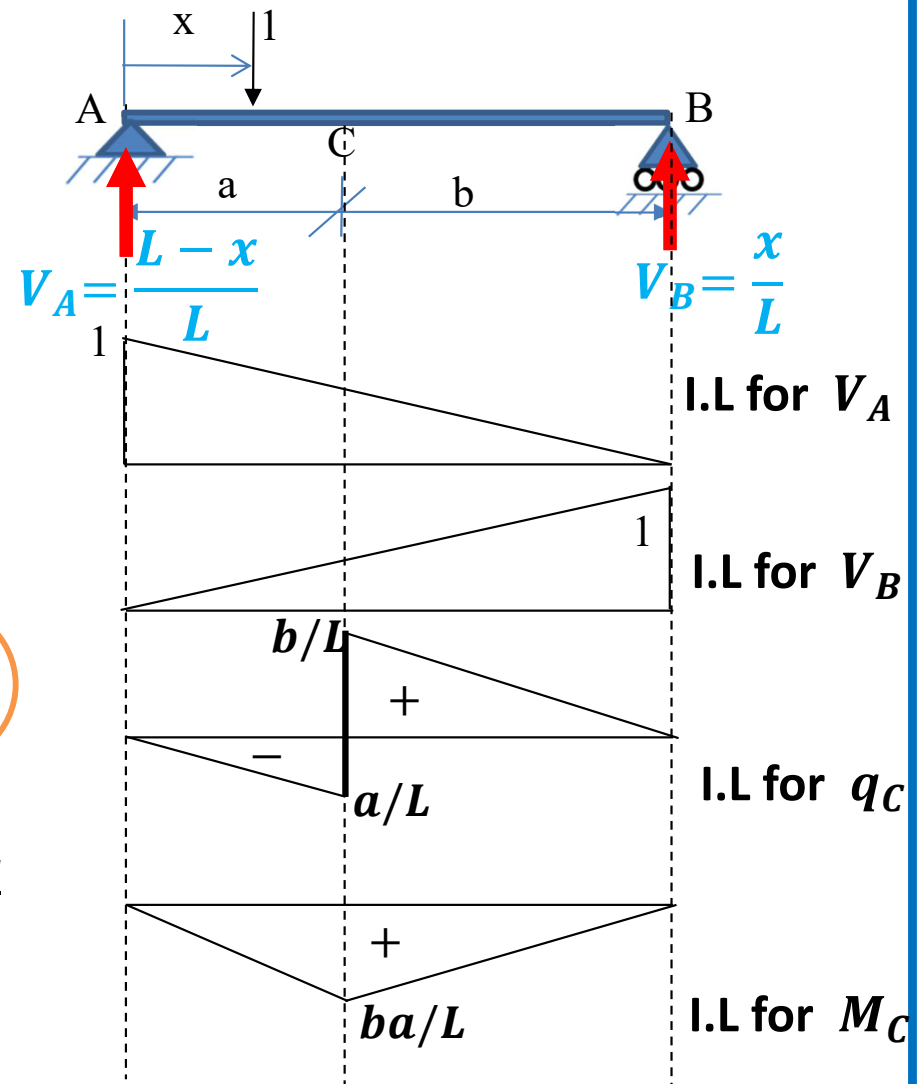
I.L for M_C :

Load on section AC:

$$\begin{aligned} M_C &= V_A a - 1(a-x) = \left(1 - \frac{x}{L}\right)a - 1(a-x) \\ &= a - \frac{ax}{L} - a + x = x - \frac{ax}{L} = \frac{Lx - ax}{L} = \frac{(L-a)x}{L} = \frac{bx}{L} \end{aligned}$$

Load on section CB:

$$M_C = V_A a = a\left(\frac{L-x}{L}\right)$$



8 INFLUENCE LINES

8.2.2 Using Muller – Breslau Principle

Muller-Breslau principle states that *the influence line for a specific load effect at a specific point is the same as the shape adopted by the structure when a unit displacement corresponding to the effect is imposed on the structure at that point, while keeping all other external forces unmoved*. [NB: The structure is considered a rigid body, therefore the adopted shape is linear, at least piece-wise].

Worked Example 8.2

Plot the I.L.D for reactions V_A and V_B , Shear at C (q_c) and Bending moment at C (M_C) for the beam shown below.

8 INFLUENCE LINES

I.L for V_A

Give V_A unit displacement vertically upwards, while keeping V_B unmoved. The resulting line is the I.L for V_A

I.L for V_B

Give V_B unit displacement vertically upwards, keeping V_A unmoved. The resulting line is the I.L for V_B

I.L for q_C

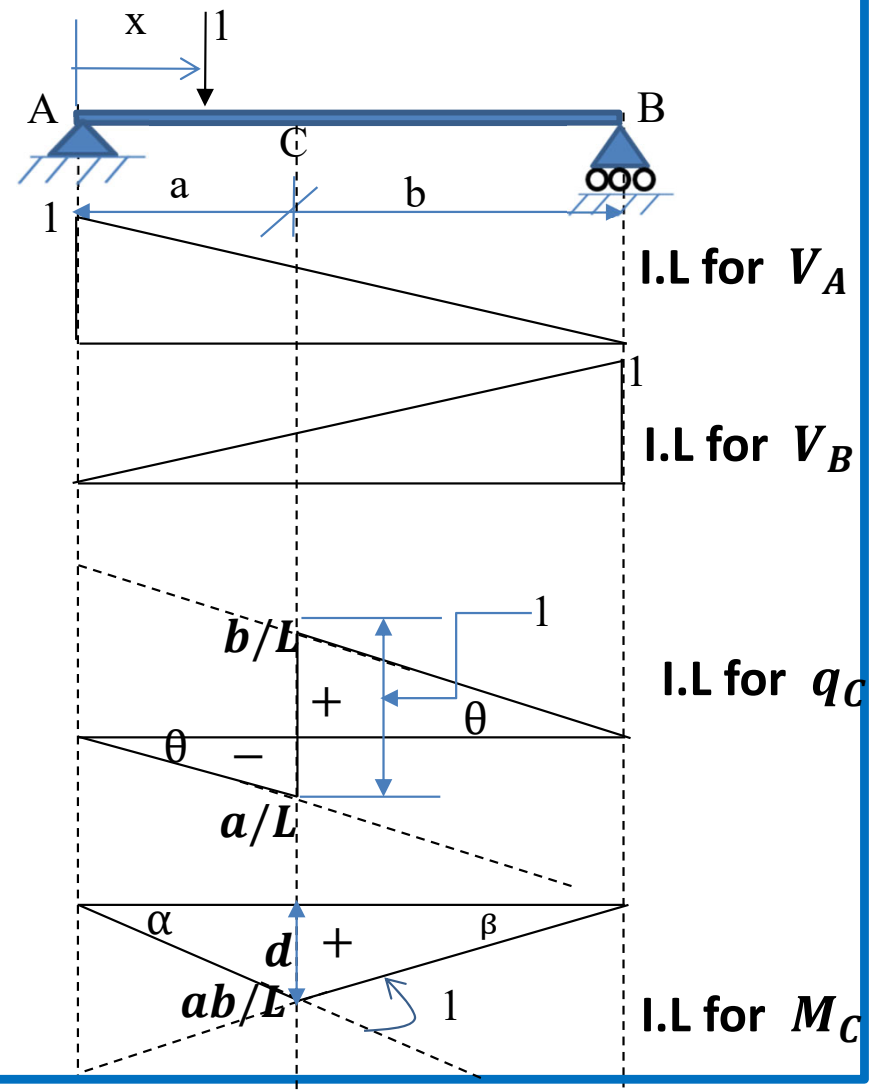
Imagine beam “cut” at C. Rotate AC and CB so that they end up being parallel (net rotation = zero), by applying a unit displacement as shown. By compatibility $\theta a + \theta b = 1 \Rightarrow \theta (a + b) = 1$
 $\Rightarrow \theta = 1/L$

Hence obtain the ordinates a/L and b/L .

I.L for M_C

Imagine beam “fractured” at C. Give CB unit rotation relative to AC by displacing point C some distance d downwards.

By compatibility: $\alpha + \beta = 1$
 $\Rightarrow (d/a + d/b) = 1 \Rightarrow d = ab/L$



8 INFLUENCE LINES

8.3 Application of Influence Lines

Influence Lines are used to determine the **load effect at a specific point**, due to a **moving load at some other point**. This is achieved as follows:

1. Point load: Load effect (at a specific point) = Magnitude of load $\times$ Influence coefficient (at that other point).
2. U.d.l: Load effect (at a specific point) = Intensity of load $\times$ Area load covers under I.L.
3. Varying load: Load effect (at a specific point) = (load function) $\times$ (I.L function) dx

Worked Example 8.3

If **a = 6m** and **b = 9m** in example 8.2, determine the maximum V_A , V_B , q_c and Bending moment M_C that

- a) A point load $P = 120 \text{ kN}$
 - b) A u.d.l. segment $w = 24 \text{ kN/m}$ of length 5 m
- would induce at the points A, B and C while traversing the beam from A to B.

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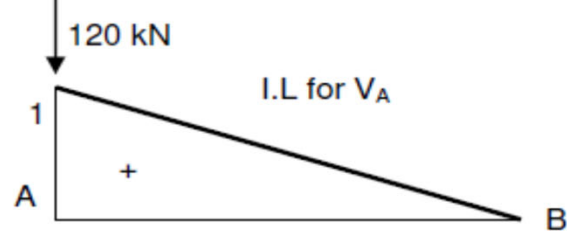
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- would induce at the points A, B and C while traversing the beam from A to B.

8 INFLUENCE LINES

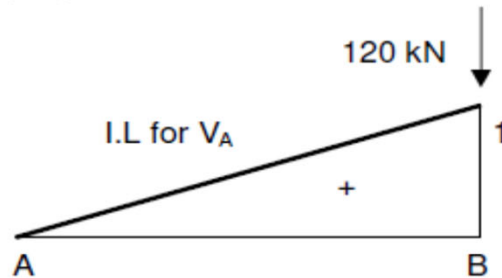
Solution

(a) Point load $P = 120 \text{ kN}$

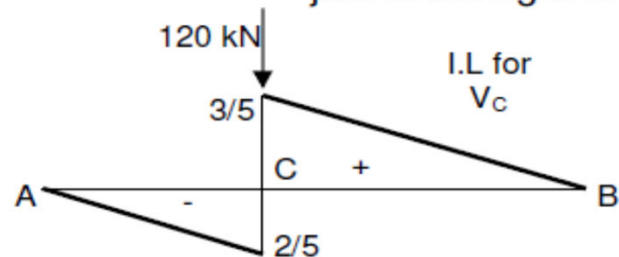
$$V_A (\text{max}) = 120 \times 1 = 120 \text{ kN}, P \text{ at A}$$



$$V_B (\text{max}) = 120 \times 1 = 120 \text{ kN}, P \text{ at B}$$

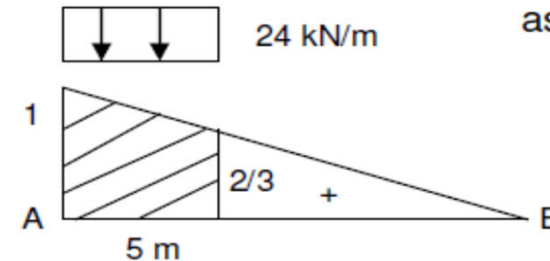


$$V_C (\text{max}) = 120 \times (3/5) = 72 \text{ kN}, P \text{ just to the right of C}$$

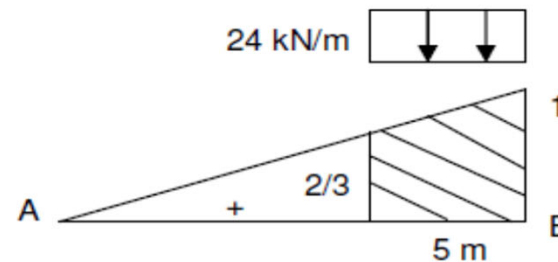


(b) U.d.I $\omega = 12 \text{ kN/m}$

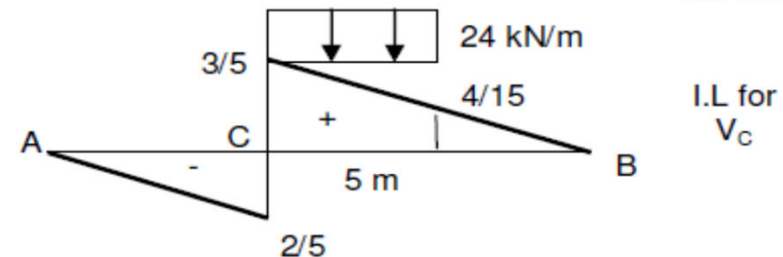
$$V_A (\text{max}) = 24(1/2)(1+2/3)(5) = 100 \text{ kN}, \text{ u.d.I arranged as shown.}$$



$$V_B (\text{max}) = 24(1/2)(2/3+1)(5) = 100 \text{ kN}, \text{ u.d.I arranged as shown.}$$



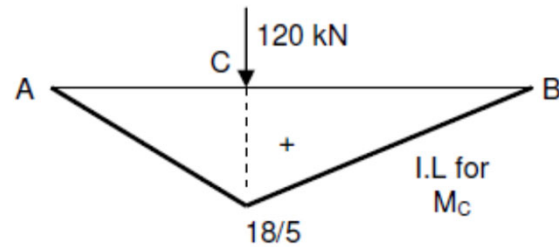
$$V_C (\text{max}) = 24(1/2)(3/5+4/15)(5) = 52 \text{ kN}, \text{ u.d.I arranged as shown}$$



8 INFLUENCE LINES

$$M_{C(\max)} = 120 \times (18/5) = 432 \text{ kNm}$$

when P arrives at C



8 INFLUENCE LINES

UDL SEGMENT SHORTER THAN SPAN

a) Maximum Shear Force

Suppose a udl segment of intensity ω and length l is to traverse a simply supported beam from A to B as shown below. Consider the cross-section X at distance x from A.

The maximum positive and maximum negative shear forces are obtained by positioning the load **strategically** so as to obtain the maximum positive and maximum negative areas under the I.L for V_X .

$$V_{X \text{ (max) (+ve)}} = (1/2)[(1 - x/L) + (1 - x/L - l/L)] \omega l$$

$$= \omega l (2L - 2x - l)/2L$$

V_X attains absolute maximum at A, when $x = 0$

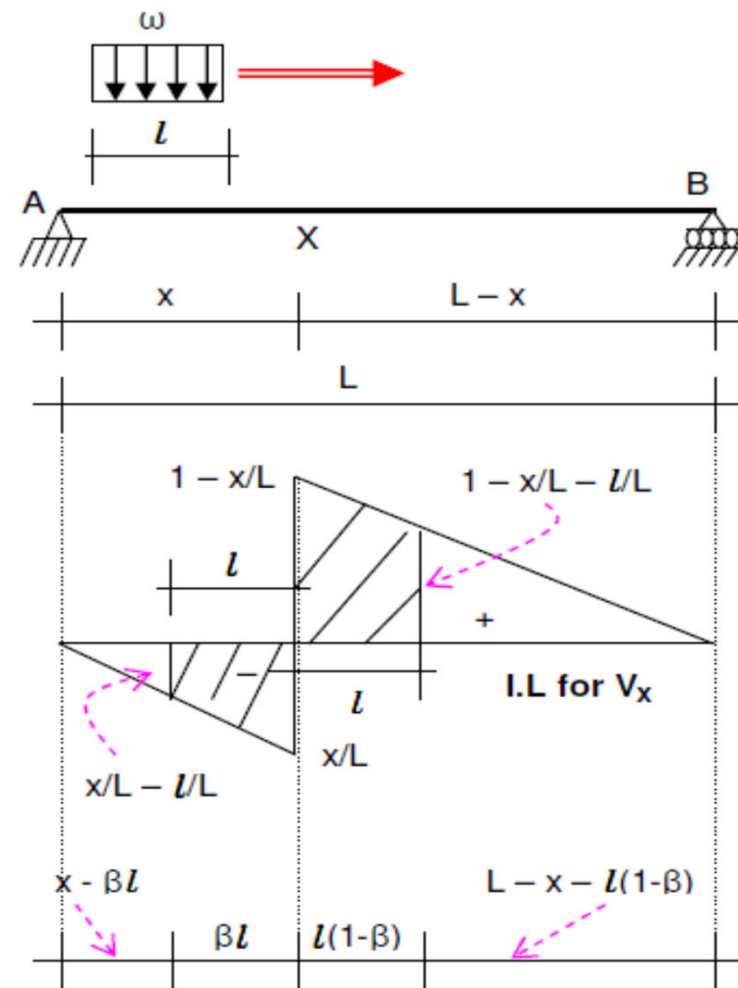
$$\text{Thus } V_{X \text{ (max) (abs) (+ve)}} = \omega l (2L - l)/2L = V_A$$

$$V_{X \text{ (max) (-ve)}} = (1/2)[(x/L - l/L) + x/L] \omega l$$

$$= \omega l (2x - l)/2L$$

V_X attains absolute maximum at B, when $x = L$

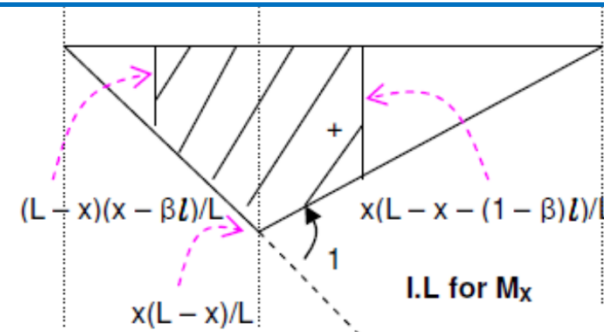
$$\text{Thus } V_{X \text{ (max) (abs) (-ve)}} = \omega l (2L - l)/2L = V_B$$



8 INFLUENCE LINES

b) Maximum Bending Moments

In order to obtain maximum moment the load must be positioned in such a manner that the area under the I. L is maximized.



Let β be the proportion of load to the left of the cross-section X, where $0 \leq \beta \leq 1$.

The shaded area is given

$$A = (1/2)\{(L-x)(x-\beta L)/L + x(L-x)/L\}\beta L + (1/2)\{x(L-x)/L + x(L-x-(1-\beta)L)/L\}(1-\beta)L$$

$$A = (1/2)\{2Lx - 2Lx^2/L - L^2x/L + 2L^2\beta x/L - L^2\beta^2\}$$

In order to obtain maximum area, set $\partial A / \partial \beta = 0 \Rightarrow 2L^2x/L - 2L^2\beta = 0 \Rightarrow \beta = x/L$

Conclusion 1: The maximum bending moment at any point occurs when that point divides the udl segment and the beam span in the same ratio.

$$\beta = \frac{\text{Load to the left of X}}{\text{Total load}} = \frac{\text{Span to the left of X}}{\text{Total span}} = \frac{x}{L}$$

The **absolute maximum bending moment** is obtained by setting $\beta = x/L$ in the expression for A, then $\partial A / \partial x = 0$. This gives $x = L/2$.

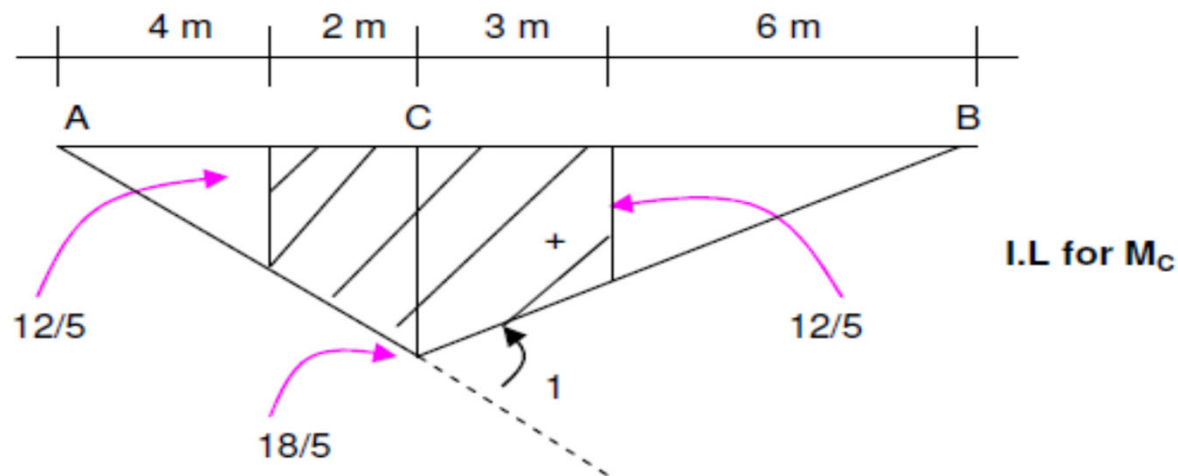
Conclusion 2: The absolute maximum moment occurs at midspan when the udl is centrally positioned there.

8 INFLUENCE LINES

Solution

$$\omega = 24 \text{ kN/m}, l = 5 \text{ m}, L = 15 \text{ m}$$

$$\text{At C, } x = 6 \text{ m} \Rightarrow \beta = 6/15 \Rightarrow \text{Load to the left of C} = 5 \text{ m} \times 6/15 = 2 \text{ m.}$$



$$\begin{aligned} \text{Shaded Area} &= (1/2) \{ (12/5 + 18/5) \times 2 + (12/5 + 18/5) \times 3 \} \\ &= (1/2) \{ (12/5 + 18/5) (2 + 3) \} \\ &= 15 \text{ m}^2 \end{aligned}$$

$$M_C (\text{max}) = 24 \times 15 = 360 \text{ kNm}$$